

Bayesian Surface Habitability Mapping of Titan

from the Late Heavy Bombardment to the Solar Red-Giant Phase

A report submitted for the degree of

Master of Philosophy

Planetary Science and Life in the Universe

Chris Meadows

cm10004@cam.ac.uk

Institute of Astronomy, University of Cambridge

March 2026

All code, processed data products, and static habitability maps are openly archived at

<https://github.com/chrimblechrumble/pslu-p3>.

1 Introduction

1.1 The Search for Habitable Environments Beyond Earth

The question of whether life exists elsewhere in the universe has undergone a profound transformation over the past three decades, shifting from the domain of philosophical speculation to empirical scientific investigation. This change has been driven by two largely independent but mutually reinforcing developments: the systematic discovery that life on Earth thrives in conditions previously considered inimical to biology, and the growing recognition through robotic spacecraft exploration that chemically and thermodynamically suitable environments for life are far more widely distributed throughout the solar system than twentieth-century intuitions suggested.

The progressive characterisation of terrestrial extremophile communities has fundamentally expanded the envelope of conditions considered compatible with life. Microorganisms have been cultured from deep-sea hydrothermal vents at temperatures exceeding 121°C, from hypersaline brines with water activities as low as $a_w = 0.60$, from the anoxic pore spaces of deep crustal rocks where electrons liberated by water–rock reactions provide metabolic energy in the total absence of sunlight, and from sub-glacial Antarctic lakes that have been isolated from the atmosphere for millions of years (Cavicchioli, 2002; Rothschild and Mancinelli, 2001). Each such discovery has moved the habitable niche boundaries outwards, making it progressively more difficult to argue that any particular planetary environment is categorically uninhabitable without specific evidence that the relevant physical and chemical conditions lie entirely outside the known envelope of life.

In parallel, the outer solar system has revealed a far richer array of liquid envi-

ronments than the classical habitable zone framework would predict. The Europa detection from the Galileo mission (Pappalardo et al., 1999) and the subsequent discovery of active cryovolcanism and ocean chemistry at Enceladus (Waite et al., 2017) established that tidal heating can maintain liquid water oceans beneath ice shells at heliocentric distances far beyond any reasonable definition of the water-based habitable zone. These discoveries shifted the conceptual framework of solar system habitability from a two-dimensional picture (terrestrial planets in a narrow orbital band) to a three-dimensional one encompassing subsurface environments throughout the outer solar system.

Within this broader astrobiological context, Saturn’s moon Titan occupies a distinctive and arguably singular position. It is the only moon in the solar system possessing a substantial atmosphere — one that is denser at the surface than Earth’s and dominated by molecular nitrogen, with a structurally complex photochemistry driven by the ultraviolet flux of the distant Sun. It is the only known extraterrestrial body to host stable surface liquids (Stofan et al., 2007), completing a full precipitation-evaporation-lake cycle that is the structural analogue of the terrestrial hydrological cycle. It may have accumulated the most complex organic chemical inventory in the solar system outside Earth itself (Lorenz et al., 2008). And at depth, it contains a water–ammonia ocean whose thermodynamic conditions are increasingly well-constrained by tidal measurements and geophysical modelling. The combination of these properties makes Titan not merely a potentially habitable world but a natural laboratory for prebiotic chemistry operating at planetary scale and across a geological timescale of ~ 4.5 Gyr.

1.2 The Physical Environment of Titan

1.2.1 Atmospheric Structure and Composition

Titan’s atmosphere was first confirmed as substantially nitrogen-rich by radio occultation measurements from the Voyager 1 flyby in November 1980, which constrained the

surface pressure to approximately 1.5 bar and the surface temperature to near 90 K (Tyler et al., 1981). The subsequent *Cassini–Huygens* mission provided a comprehensive characterisation that transformed our understanding of atmospheric structure. The Huygens Atmospheric Structure Instrument (HASI) measured a surface pressure of 1.467 ± 0.001 bar and a surface temperature of 93.65 ± 0.25 K at the probe’s landing site (192.3°W , 10.6°S), with a tropospheric lapse rate of ~ 1.38 K km $^{-1}$ and a tropopause cold trap at ~ 40 km altitude where the temperature reaches a minimum of ~ 70 K (Fulchignoni et al., 2005). A stratospheric temperature inversion produces a warmer mesosphere above, analogous to the terrestrial stratosphere maintained by ozone absorption.

The *Huygens* Gas Chromatograph Mass Spectrometer (GCMS) measured a surface composition of $\approx 94.2\%$ molecular nitrogen (N_2), $5.65 \pm 0.18\%$ methane (CH_4), trace argon-40 (≈ 43 ppm, consistent with outgassing of ^{40}K from the interior), and a variety of simple organic molecules including cyanogen, acetonitrile, and propionitrile (Niemann et al., 2005). At higher altitudes, the Cassini INMS and CAPS instruments revealed a far richer chemistry: complex nitriles, benzene, and high-molecular-weight organic ions characteristic of nascent aerosol formation are detected above ~ 900 km, confirming that the tholins production process begins in the thermosphere and mesosphere and proceeds downward through the stratosphere to the surface (Waite et al., 2007).

The methane cycle in Titan’s troposphere constitutes the dominant agent of surface change and chemical transport. Methane evaporates from the north polar hydrocarbon seas, is transported equatorward by the global Hadley circulation, condenses and precipitates as rainfall over equatorial and mid-latitude terrain (Turtle et al., 2011a), and drains back towards polar basins through an extensive network of fluvial channels that Miller et al. (2021) have mapped globally from Cassini SAR data. The cycle operates on timescales ranging from individual storm events of hours to decades, through the 29.5-year Saturnian seasonal cycle, to longer-period lake-level variations driven by orbital forcing. Its significance for habitability lies in the continuous deliv-

ery of organic material from photochemically active equatorial regions to the polar lake margins, and in the chemical gradients and concentration cycles it produces at evaporating shorelines.

Ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere produces an extraordinary variety of complex organic photochemical products collectively termed tholins (Sagan and Khare, 1979). Laboratory synthesis of tholins from N_2/CH_4 mixtures under UV irradiation produces hundreds of identifiable molecular species including simple nitriles, amino acid precursors, nucleobase analogues, and sugar-like polyols (Cable et al., 2012). The production rate of organic aerosols is estimated at $\sim 5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$ (Lavvas et al., 2011), implying that the surface has accumulated an organic layer estimated at between 1 and 10 km depth over solar system history (Lorenz et al., 2008). This organic reservoir is the primary chemical substrate for all habitability scenarios considered in this thesis, whether in the present epoch (as feedstock for acetylenotrophy), in the geologically recent past (as reactant in impact-melt amino acid synthesis), or in the far future (as dissolved substrate in the red-giant water–ammonia ocean).

1.2.2 Surface Geology and Geomorphology

The Cassini RADAR instrument mapped approximately 67% of Titan’s surface in synthetic aperture radar (SAR) mode at $\sim 350 \text{ m}$ per pixel resolution over 127 targeted flybys spanning 2004–2017. This coverage, whilst substantial, leaves approximately one-third of the surface — concentrated in mid-southern latitudes and portions of the northern hemisphere — dependent on lower-sensitivity ISS and VIMS imagery alone. The SAR dataset revealed a geologically diverse world operating under conditions profoundly different from any other solar system body: a geology dominated not by silicate tectonism or basaltic volcanism but by organic sediment transport, cryovolcanism, and the weathering and dissolution of ice by liquid hydrocarbon.

Lopes et al. (2019) synthesised the full Cassini SAR dataset into the first comprehensive global geomorphological map of Titan, identifying seven terrain classes that

account for all mapped surface morphologies: plains ($\sim 65\%$ of the SAR-imaged surface), dune fields ($\sim 17\%$), mountains and hummocky terrain ($\sim 8\%$), basins ($\sim 3\%$), labyrinth terrain ($\sim 4\%$), impact craters ($< 1\%$), and polar hydrocarbon lakes and seas ($\sim 1.5\%$) (Malaska et al., 2025). This geomorphological map forms a cornerstone of the habitability framework developed in this thesis, both as a source of terrain classification information for the organic abundance feature and as context for interpreting the spatial patterns in the posterior habitability maps.

The equatorial dune seas — Shangri-La, Belet, Aztlan, and Fensal, together forming a near-continuous band from roughly -30 to $+30$ latitude — are the single largest geomorphological unit on the planet. These linear dunes, oriented consistent with eastward net sand transport by the atmospheric circulation (Tokano, 2010), are composed of organic particulates, most likely complex hydrocarbons and nitriles derived from atmospheric tholin deposition (Lorenz et al., 2006). Near-infrared spectroscopy from VIMS confirms that dune terrain has the highest $1.59/1.27\text{ }\mu\text{m}$ band ratio of any terrain class on Titan, indicative of the highest tholin content (Le Mouélic et al., 2019). The dune belt accordingly receives the highest organic abundance score (0.82) in the geomorphological classification used here (Appendix C).

The north polar region above $\sim 60^\circ\text{N}$ hosts the three major hydrocarbon seas: Kraken Mare (area $\sim 400,000\text{ km}^2$, estimated maximum depth $\sim 300\text{ m}$), Ligeia Mare ($\sim 126,000\text{ km}^2$, depth $\sim 200\text{ m}$), and Punga Mare ($\sim 38,000\text{ km}^2$), together containing an estimated $\sim 70,000\text{ km}^3$ of liquid (Mastrogiuseppe et al., 2019). The compositions of the two largest seas differ measurably. Cassini active radar depth sounding and microwave emissivity measurements indicate that Ligeia Mare is predominantly methane-rich, whilst Kraken Mare is more ethane-rich (estimated at approximately 70% ethane, 16% propane, 7% methane, and 7% nitrogen by mole fraction near the 2004-era shoreline position), with composition varying seasonally as methane preferentially evaporates (Mastrogiuseppe et al., 2019). These chemical differences between adjacent sea bodies imply distinct chemical environments at their respective shorelines and will be reflected in the habitability mapping.

The Xanadu region ($\sim 90\text{--}150^\circ\text{W}$, equatorial) is anomalously radar-bright in SAR and spectrally dominated by crystalline water ice in VIMS measurements, in strong contrast to the organic-dominated signature of surrounding terrain (Brown et al., 2011). Its origin remains debated: proposed explanations include excavation of water-ice-rich crustal material by a large ancient impact (Radebaugh et al., 2011), a relict tectonic province, or anomalous ice deposition. Whatever its origin, Xanadu’s water-ice-rich composition results in a low organic abundance score and consequently low habitability score in the present model.

Ontario Lacus at the south pole (179°W , 72°S) is the only confirmed southern-hemisphere lake (Turtle et al., 2009), with an estimated area of $\sim 15,000\text{ km}^2$. ISS observations revealed measurable shoreline recession between 2005 and 2010, suggesting net evaporation under southern summer insolation at the Cassini epoch. Its significantly smaller size and lower associated methane cycle activity compared to the north polar seas means it scores notably lower in the present-epoch habitability map.

1.2.3 Interior Structure and the Subsurface Ocean

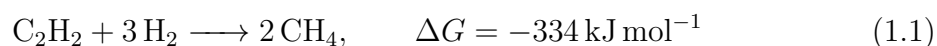
The most critical interior constraint for habitability purposes is the tidal Love number $k_2 = 0.589 \pm 0.150$ measured by Iess et al. (2012) from orbital perturbations of the Cassini spacecraft during close flybys. This value greatly exceeds the prediction for a fully solid (rocky core plus solid ice shell) body, for which $k_2 < 0.03$. The only viable physical explanation is a global subsurface liquid layer that deforms readily under Saturn’s tidal force, amplifying the tidal response. Geophysical interior models constrain this layer to a water–ammonia ocean at $\sim 55\text{--}80\text{ km}$ depth, beneath an outer ice-I shell and overlying an ice-VI mantle and silicate core (Nimmo and Bills, 2010; Tobie et al., 2005). The ocean’s ammonia concentration is estimated at 5–15 wt% from density constraints (Mitri et al., 2014); this concentration depresses the melting point of the mixture to near the water–ammonia eutectic (176 K at 32.6 wt% NH_3 ; Grasset and Pargamin 2005), ensuring the ocean remains liquid at its current depth despite surface temperatures of only 94 K.

The subsurface ocean has direct habitability relevance for two reasons. First, [Affholder et al. \(2025\)](#) demonstrated that glycine fermentation ($\Delta G \approx -20$ to -50 kJ/mol under estimated ocean conditions) is energetically viable within the ocean, and that the dissolved organic nitrogen available from hydrolysed tholins provides adequate substrate for simple metabolic cycles. Second, at the surface, SAR-bright annular features surrounding several large impact craters are interpreted as ancient melt-water fronts where liquid water sourced from the subsurface ocean (or from impact melting) spread outward from crater basins, potentially bringing ocean-derived chemistry into contact with the surface organic inventory ([Wood et al., 2010](#)). This interpretation underpins the elevated subsurface ocean proximity score at Selk crater discussed in Chapter 3.

1.3 Chemical Habitability of Titan’s Surface: Prior Theoretical Work

1.3.1 The Acetylenotrophy Hypothesis

The first quantitative assessment of the energetic viability of life on Titan’s surface was provided by [McKay and Smith \(2005\)](#), who evaluated the metabolic energy available from the combination of atmospheric acetylene (C_2H_2) with molecular hydrogen (H_2) — a process they termed acetylenotrophy:



Acetylene is produced continuously by ultraviolet photolysis of methane in the upper atmosphere and sediments to the surface. Molecular hydrogen, a photochemical by-product, similarly settles downward. McKay and Smith calculated that the available acetylenotrophy energy flux at Titan’s surface is comparable to that available from photosynthesis at the deep ocean floor on Earth, and substantially exceeds the metabolic energy requirements of the most energy-sparse known terrestrial organisms,

including methanogens and anaerobic chemoautotrophs.

Possible observational support for this hypothesis emerged from [Strobel \(2010\)](#), who reported a downward flux of H_2 to Titan's surface of $\sim 10^{10} \text{ molecules cm}^{-2} \text{ s}^{-1}$ — approximately 100 times the flux predicted by known abiotic surface reactions — based on analysis of Cassini INMS data. A surface sink for H_2 exceeding abiotic predictions is precisely consistent with acetylenotrophy. However, [Teolis et al. \(2015\)](#) subsequently identified INMS calibration uncertainties at low signal levels that could affect the reported flux magnitude, and the measurement must be considered tentative pending confirmation by future in-situ measurements. The energetics of acetylenotrophy were revisited using updated thermochemical data by [Yanez et al. \(2024\)](#), who calculated a free energy yield of $69\text{--}78 \text{ kJ mol}^{-1} \text{ C}$ under Titan surface conditions — somewhat more conservative than the original McKay & Smith estimate but still comfortably above the minimum for terrestrial methanogens ($\sim 25\text{--}65 \text{ kJ mol}^{-1} \text{ C}$).

1.3.2 Non-Aqueous Biochemistry and the Azotosome Hypothesis

A complementary line of theoretical work has examined the possibility of life operating with liquid methane as its biochemical solvent rather than water. [Benner et al. \(2004\)](#) provided a theoretical basis for non-aqueous biochemistry, arguing that whilst water's polarity and hydrogen-bonding capacity are indeed important for terrestrial biochemistry, these properties are not thermodynamically fundamental requirements; a sufficiently different but internally consistent biochemistry might evolve under different solvent conditions. In liquid methane at $\sim 90 \text{ K}$, the principal structural challenge is membrane formation: conventional phospholipid bilayers, whose amphiphilic nature depends on the polarity contrast between water and non-polar lipid tails, would not form spontaneously in a non-polar solvent.

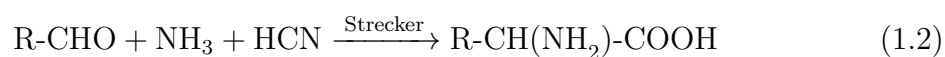
[Stevenson et al. \(2015\)](#) addressed this challenge through molecular dynamics simulations, showing that small nitrogen-containing molecules such as acrylonitrile (CH_2CHCN), subsequently detected in Titan's stratosphere by ALMA ([Palmer et al., 2017](#)), can

spontaneously form stable membrane-like vesicles (termed azotosomes) in liquid methane. These azotosome membranes would have reversed polarity relative to terrestrial cell membranes, with polar head groups directed inward towards a polar interior. Whilst no experimental demonstration of azotosome-based metabolism exists, the thermodynamic and structural feasibility of membrane formation establishes at least one plausible path to cellular compartmentalisation in a methane-solvent biology.

1.3.3 Prebiotic Chemistry in Post-Impact Melt Environments

A third and distinctly different habitability pathway on Titan involves transient liquid water rather than the permanent hydrocarbon lakes. When large impactors strike Titan’s ice-rich crust, the kinetic energy of impact melts the surrounding material, producing water–ammonia–organic mixture ponds that persist at liquid temperatures for 10^3 – 10^4 years before refreezing (O’Brien et al., 2005). For a 50 km diameter rocky impactor at Titan’s escape velocity of 2.64 km s^{-1} , the delivered kinetic energy is sufficient to melt $\sim 10^4 \text{ km}^3$ of ice. During the Late Heavy Bombardment the impactor flux was approximately 100–1000 times the present background rate, ensuring that multiple active melt pools existed simultaneously on Titan’s surface at any time during the $\sim 300 \text{ Myr}$ duration of the bombardment.

Madan et al. (2026) demonstrated through thermodynamic and kinetic modelling that the Strecker reaction:



is accessible in post-impact melt environments at temperatures of 200–350 K. All three precursors are available: surface aldehydes (R-CHO) from tholin decomposition, NH_3 from the water–ammonia melt, and HCN from the nitrogen-rich atmosphere (HCN is one of the most abundant photochemical products in Titan’s stratosphere; Vuitton et al. 2007). The amino acid products are stable on timescales comparable to the melt pool lifetime. This pathway is central to the astrobiological rationale for the

Dragonfly mission’s primary landing site at Selk crater, where the post-impact melt chemistry of an ancient large impact is preserved in the geological record.

1.3.4 The Far-Future Habitability Window

Lorenz et al. (1997) identified a fourth temporal window of classical water-based habitability on Titan associated with the Sun’s eventual departure from the main sequence. As solar luminosity increases by factors of hundreds over $\sim 10^7$ – 10^8 yr during the red-giant phase, Titan’s surface temperature will rise through the water–ammonia eutectic (176 K) and remain above it for an estimated 200–500 Myr (Lorenz et al., 1997; Schröder and Connon Smith, 2008). During this window, Titan’s entire surface would be covered by a global water–ammonia ocean, overlying an organic sediment reservoir that by that epoch will have accumulated for ~ 16 Gyr of continuous atmospheric deposition — a chemical stockpile of extraordinary richness. In this scenario, Titan would become transiently and potentially uniquely the most habitable body in the solar system, at a time when Earth and Mars will have been sterilised by the expanding solar photosphere. The quantitative characterisation of this future epoch is one of the novel contributions of this thesis.

1.4 The Cassini–Huygens Mission and Its Astrobiological Legacy

The joint NASA/ESA *Cassini–Huygens* mission operated in the Saturnian system from July 2004 until September 2017, when the spacecraft was deliberately deorbited into Saturn’s atmosphere to prevent biological contamination of potentially habitable moons. The 13-year operational lifetime enabled 127 targeted Titan flybys, the *Huygens* probe descent and landing in January 2005, and a systematic multi-instrument survey of Titan’s atmosphere, surface, and interior that constitutes the observational foundation of this thesis.

The observational coverage is comprehensive but incomplete. SAR mapping reached

~67% of the surface; the VIMS near-infrared spectral coverage relevant to organic abundance assessment reaches approximately 50%; the CIRS thermal mapping achieved global coverage at ~10° latitude resolution; and the radar altimetry dataset covers ~90% of the globe after global interpolation. The GCMS aboard *Huygens* provided the single direct measurement of surface chemical composition, at one equatorial site. This pattern — broad but spatially uneven remote sensing coverage combined with a single point of in-situ ground truth — defines both the strengths and the principal limitations of the quantitative habitability framework developed here.

The post-*Cassini* synthesis volumes *Titan After Cassini-Huygens* (Cambridge University Press, 2025) have distilled the scientific legacy of the mission. In the context of surface habitability, the synthesis highlights three significant gaps: (1) the absence of any quantitative spatially-resolved habitability probability map of the full globe; (2) the absence of an explicit multi-epoch temporal treatment connecting the LHB, present, and far-future habitability regimes; and (3) the absence of a reproducible, openly archived analysis framework integrating the full suite of Cassini data modalities. This thesis addresses all three.

1.5 The Dragonfly Mission

NASA’s *Dragonfly* mission (Lorenz et al., 2018) was selected as the fourth New Frontiers programme mission in June 2019. It is currently en route to Titan with a projected arrival in 2034. The mission deploys a dual-quadcopter rotorcraft that exploits Titan’s dense atmosphere (surface density $\rho \approx 5.4 \text{ kg m}^{-3}$) and low gravity to achieve powered flight, enabling mobility impossible for wheeled rovers. The baseline mission plans for ~175 km of total traverse distance over a 2.7-year nominal mission, beginning in the Belet dune field to the north-west of Selk crater and progressing through the crater ejecta blanket to the crater floor.

The scientific rationale for Selk as the primary target rests on the exposure of water-ice-rich crustal material in the ejecta (enabling Strecker amino acid synthesis when contacted by melt water), the adjacent organic-rich dune terrain, and the radar-

bright crater rim morphology consistent with ancient liquid-water exposure. *Dragonfly* carries a mass spectrometer (DraMS), a gamma-ray and neutron spectrometer (DraGNS), a drill, a camera suite (DragonCam), and a meteorology package. The analysis in this thesis provides a quantitative evaluation of the mission’s site selection and identifies which habitability features each instrument is best positioned to test.

1.6 Scope and Research Questions

Previous habitability assessments of Titan have shared four important limitations. They are predominantly qualitative or point-specific, lacking global spatial resolution. They treat the present epoch as the only relevant state, ignoring both the LHB prebiotic-chemistry window and the far-future red-giant window. They do not integrate the full suite of Cassini data modalities into a unified probabilistic framework. And they lack the reproducibility afforded by an openly archived analysis pipeline.

The closest methodological precedent is [Affholder et al. \(2021\)](#), who applied Bayesian statistical inference to Cassini INMS plume data from Enceladus, comparing the observed methane and hydrogen escape rates against models of abiotic serpentinisation and biotic methanogenesis. Their finding — that the observed methane escape rate is most parsimoniously explained by biological methanogenesis if the prior probability of life is sufficient — represents the most quantitatively rigorous Bayesian biosignature assessment yet published for a solar system body. The present work is directly inspired by this approach, extended from a single-point plume observation to a full-globe, multi-feature, multi-epoch spatial mapping.

The specific research questions addressed are:

RQ1 *What is the spatial distribution of surface habitability on present-day Titan, and which terrain types, geological features, and atmospheric zones produce the highest scores? Are the data-derived feature importances consistent with the physically motivated prior weights?*

RQ2 *How did habitability differ during the Late Heavy Bombardment, when impact-*

melt ponds provided transient liquid water? Which impact structures were the most astrobiologically significant, and what are the implications for prebiotic chemistry?

RQ3 *How does habitability evolve through the temporal sequence from the LHB to the red-giant future? Is Titan potentially the most habitable body in the solar system during the red-giant ocean window?*

RQ4 *Does the derived habitability map independently validate the Dragonfly landing site at Selk crater, and which instruments are best positioned to test the model's predictions?*

Chapter 2 presents the data sources, feature extraction, Bayesian inference model, and temporal methodology. Chapter 3 presents and interprets the habitability maps at each epoch, the feature importances, the temporal evolution, and the detailed Selk crater analysis. Chapter 4 discusses the *Dragonfly* mission implications, biosignature framework, limitations, and future work.

2 Methods

2.1 Overview of the Analytical Approach

2.1.1 Acronyms and Nomenclature

The following acronyms are used throughout this chapter and the remainder of the thesis. They are defined here for convenience and also at the point of first use.

SAR	Synthetic Aperture Radar. An active microwave imaging technique in which the motion of the spacecraft is used to synthesise a very long antenna aperture, yielding fine-resolution imagery of surface texture and composition.
VIMS	Visible and Infrared Mapping Spectrometer. The <i>Cassini</i> instrument that imaged Titan’s surface through six near-infrared atmospheric transparency windows, providing surface composition data.
CIRS	Composite Infrared Spectrometer. The <i>Cassini</i> thermal emission instrument used here to constrain the surface temperature field.
GTDR	Global Titan Digital elevation model, Radar — the raw Cassini RADAR altimetry dataset archived in the Planetary Data System.
GTDE	Global Titan Digital Elevation model — the globally interpolated topographic product of Corlies et al. (2017) derived from the GTDR and supplementary data sources.
DEM	Digital Elevation Model. A gridded representation of surface topography.

PDS	Planetary Data System. The NASA archive for planetary mission data.
HDI	Highest Density Interval. A Bayesian credible interval containing the specified probability mass within the narrowest possible range.

2.1.2 Notation: Feature Weights and Prior Means

Two quantities appear throughout this chapter and are defined here to avoid repetition. Each habitability-proxy feature f_i ($i = 1, \dots, 8$) is assigned:

- A **weight** $w_i \in [0, 1]$, where $\sum_i w_i = 1$. The weight encodes the relative astrobiological importance of feature i as a habitability indicator. A feature with a higher weight exerts a larger influence on the posterior probability of habitability, all else being equal. Weights are assigned from physical reasoning and validated against data-derived importances (Section 3.2.5).
- A **prior mean** $\mu_i \in [0, 1]$. This is the expected spatially-averaged value of feature i before any observational data are considered — a statement of our prior knowledge about how common or widespread the habitability-relevant condition is on Titan. For example, liquid hydrocarbon covers $\sim 2\%$ of Titan’s surface (Hayes et al., 2008), so its prior mean is $\mu_1 = 0.02$. The weighted sum of all prior means defines the global prior mean habitability: $\mu_0 = \sum_i w_i \mu_i = 0.331$. Note that the value $\mu_0 = 0.331$ does not appear in the prior literature as a global habitability estimate for Titan; it emerges from the component prior means in Table B.1 and should be interpreted as encoding the observation that Titan’s surface is globally organic-rich whilst simultaneously recognising that liquid solvent is present at only $\sim 2\%$ of the surface. The resulting moderate prior is intentionally revisionable: the weak concentration parameter $\kappa = 5$ ensures that the data, not the prior, determine the posterior at any pixel where observational coverage is good. Full values are tabulated in Appendix B.

This notation first appears in a practical context in Section 2.3, where the eight features are described individually. Their role in the Bayesian update is explained in Section 2.4.

2.1.3 The Bayesian Approach: Motivation and Structure

The central methodological challenge in quantifying Titan’s surface habitability is that no single Cassini instrument directly measures habitability. Each instrument provides a partial and indirect proxy: radar backscatter reveals surface composition and liquid surfaces; near-infrared spectroscopy constrains organic content; thermal emission maps the temperature field that drives the methane cycle; altimetry constrains topography and hence organic accumulation patterns; and the geomorphological classification synthesises the collective information from all instruments into terrain types whose habitability implications can be grounded in laboratory and theoretical work.

The approach taken here is to combine these partial proxies through a Bayesian inference framework that produces, at each grid cell, a posterior probability of habitability $P(H \mid \mathbf{f})$ given the vector of observed features $\mathbf{f} = (f_1, \dots, f_8)$. This is structurally analogous to the approach of [Affholder et al. \(2021\)](#), who used Bayesian methods to evaluate whether the observed methane flux in Enceladus’s plumes is more consistent with abiotic serpentinisation or with biological methanogenesis. The key extensions here are spatial resolution (a 6.49-million-pixel global map rather than a single-point estimate), the number of independent observational inputs (eight features rather than one), and the temporal scope (five epochs spanning geological time rather than a single snapshot).

The structure of the inference is: (1) encode prior knowledge about Titan’s habitability into a probability distribution over the latent variable H ; (2) update that distribution using each observed feature f_i as evidence; and (3) report the resulting posterior probability $P(H \mid \mathbf{f})$ as the habitability estimate at each pixel. The mathematical details of this procedure are given in Section 2.4; the present section focuses on the observational inputs.

The analysis is applied at 4490 m/px resolution on a canonical equirectangular grid covering the full globe (1802 pixels in latitude $\times$ 3603 pixels in longitude), using a west-positive longitude convention consistent with all Cassini raster products. This resolution is chosen to match the coarsest primary dataset (the GTDR altimetry mosaic at ~ 22 km per pixel) whilst resolving lake margins and fluvial features at ~ 45 km scale. All eight input datasets are reprojected to this common grid before feature extraction.

Figure 2.1 provides a schematic overview of the five-stage analytical pipeline: data acquisition, preprocessing to the canonical grid, feature extraction, Bayesian inference, and visualisation.

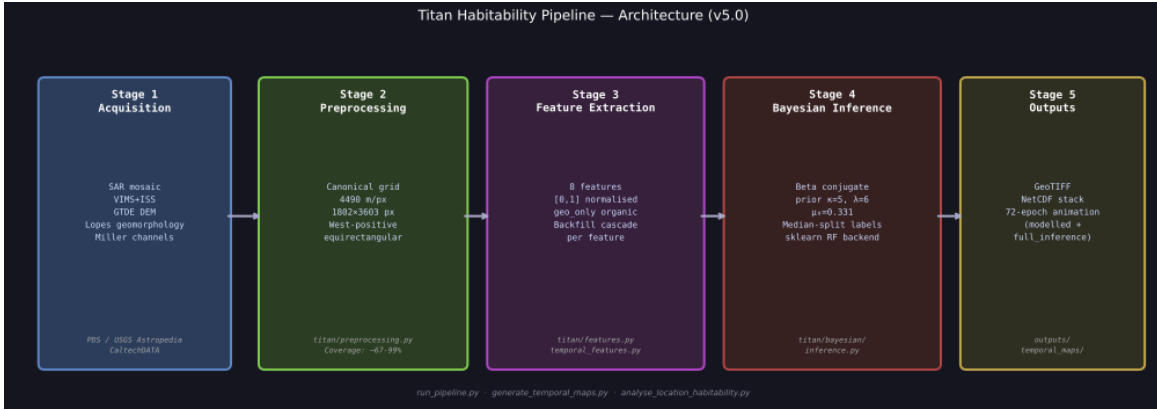


Figure 2.1: Schematic overview of the five-stage Titan habitability analysis pipeline. All eight habitability-proxy features are extracted in Stage 3 and combined in the Bayesian posterior update of Stage 4 to produce $P(H | \mathbf{f})$ maps at each of the five temporal configurations.

2.2 Observational Data Sources

2.2.1 Cassini RADAR Synthetic Aperture Radar Mosaic

The Cassini RADAR instrument operated in SAR mode during Titan flybys, imaging the surface at ~ 350 m per pixel over the 127 targeted flybys from 2004 to 2017 (Janssen et al., 2016). SAR measures the normalised radar cross-section σ^0 (typically expressed in dB), which is sensitive to surface roughness, dielectric properties, and the presence of smooth or conducting materials. Low backscatter ($\sigma^0 < -20$ dB) is

diagnostic of smooth surfaces (consistent with liquid hydrocarbon lakes) or dielectrically absorbing material (consistent with tholin-rich dune sand). High backscatter correlates with rough, water-ice-rich material such as mountain terrain, crater ejecta, and some cryovolcanic surfaces. The SAR mosaic covers $\sim 67\%$ of Titan’s surface and constitutes the primary data source for the liquid hydrocarbon, chemical energy, and subsurface ocean features.

2.2.2 VIMS Near-Infrared Spectral Mosaic

The Cassini Visible and Infrared Mapping Spectrometer (VIMS) observed Titan’s surface through atmospheric transparency windows at 0.94, 1.08, 1.27, 1.59, 2.03, and 2.79 μm . The band ratio 1.59/1.27 μm is an established proxy for tholin (complex organic) abundance at the surface: the 1.59 μm window probes deeper into the tholin-bearing surface whilst the 1.27 μm window provides atmospheric normalisation (Barnes et al., 2007). The Seignovert et al. (2019) combined VIMS+ISS mosaic (Seignovert et al., 2019) provides this band ratio map at 2–4 km per pixel coverage, but only for the 0–180°W hemisphere ($\sim 50\%$ of the globe). The significance of this coverage gap for the organic abundance feature is discussed in Section 2.3.2.

2.2.3 CIRS Thermal Emission Observations

The Composite Infrared Spectrometer (CIRS) measured Titan’s surface brightness temperature at far-infrared wavelengths throughout the Cassini mission. Jennings et al. (2019) derived an analytical model of the latitude-dependent surface temperature from 13 years of CIRS observations, producing the formula:

$$T(\phi, Y) = (93.53 - 0.095 Y) \cos[(\phi + 0.85 - 3.2 Y)(0.0029 - 0.00006 Y)] \quad (2.1)$$

where ϕ is latitude in degrees and $Y = t_{\text{CE}} - 2009.61$ is years from the northern spring equinox (22 May 2009). This formula reproduces the observed zonal surface temperatures to within ± 0.4 K (rms) across the full Cassini epoch and provides 100%

global coverage. It is used to synthesise the global temperature field and its meridional gradient, which enters the methane cycle feature (Section 2.3.4).

2.2.4 GTDE/GTDR Topographic Models

The Global Titan Digital Elevation model (GTDE), produced by [Corlies et al. \(2017\)](#), integrates Cassini RADAR altimetry, SARtopo stereophotogrammetry, and radial basis function interpolation to produce a globally-interpolated topographic model at ~ 22 km per pixel (2 pixels per degree). Coverage is approximately 90% of the globe, with the south polar region and some high-latitude northern areas relying most heavily on interpolation. The topographic model contributes to the chemical energy feature (identifying topographic lows as preferential organic accumulation sites), the topographic complexity feature (through local roughness estimation), and the surface–atmosphere interaction feature (through slope calculation).

2.2.5 Lopes (2019) Global Geomorphological Map

The global geomorphological map of [Lopes et al. \(2019\)](#) classifies the Cassini SAR-imaged surface into seven terrain units (plains, dunes, mountains, basins, labyrinth terrain, craters, and lakes) using SAR texture, backscatter, morphology, and VIMS spectral correlation. This map provides the terrain classification that enters the organic abundance feature (as a globally seamless gap-fill) and the geomorphological diversity feature. The shapefiles are provided in east-positive geographic coordinates and require conversion to the west-positive convention of the analysis grid.

2.2.6 Miller (2021) Global Fluvial Channel Map

[Miller et al. \(2021\)](#) identified and mapped Titan’s global fluvial channel network from Cassini SAR data, producing a vector dataset of channel locations and widths. This network constitutes the primary mechanism for organic material transport from equatorial source regions to the polar basins, and channel density is used as one component of the surface–atmosphere interaction feature.

2.2.7 Tidal Love Number and Gravity Data

The degree-2 tidal Love number $k_2 = 0.589 \pm 0.150$ from [Iess et al. \(2012\)](#) provides a global constraint on interior structure, confirming the presence of a subsurface liquid layer. This scalar quantity enters as a global floor value in the subsurface ocean proximity feature.

2.2.8 Hedgepeth (2020) Global Crater Catalogue

The global catalogue of confirmed and probable impact structures on Titan by [Hedgepeth et al. \(2020\)](#) records crater diameters and positions for all resolved impact features in the Cassini SAR dataset. This catalogue is used exclusively in the PAST epoch to generate the impact-melt proximity feature.

2.3 Feature Extraction and Scientific Basis

Eight habitability-proxy features are extracted from the above data sources. Figure 2.2 shows all eight feature maps at the present epoch, illustrating their distinct spatial patterns. Figure 2.3 shows how the relative activity of each feature is modulated across geological time.

All features are normalised to the range $[0, 1]$ using a robust percentile-clipping approach:

$$\hat{f}_i = \text{clip}\left(\frac{f_i - f_{i,p2}}{f_{i,p98} - f_{i,p2}}, 0, 1\right) \quad (2.2)$$

where $f_{i,p2}$ and $f_{i,p98}$ are the 2nd and 98th percentiles of finite values in the global layer, mapping the central 96% of the observed dynamic range onto $[0, 1]$ whilst preventing extreme outliers from compressing the scale. Where a feature is unavailable at a particular pixel (due to data gaps), the value is set to NaN and handled through the Bayesian missing-data imputation procedure described in Section 2.4.5.

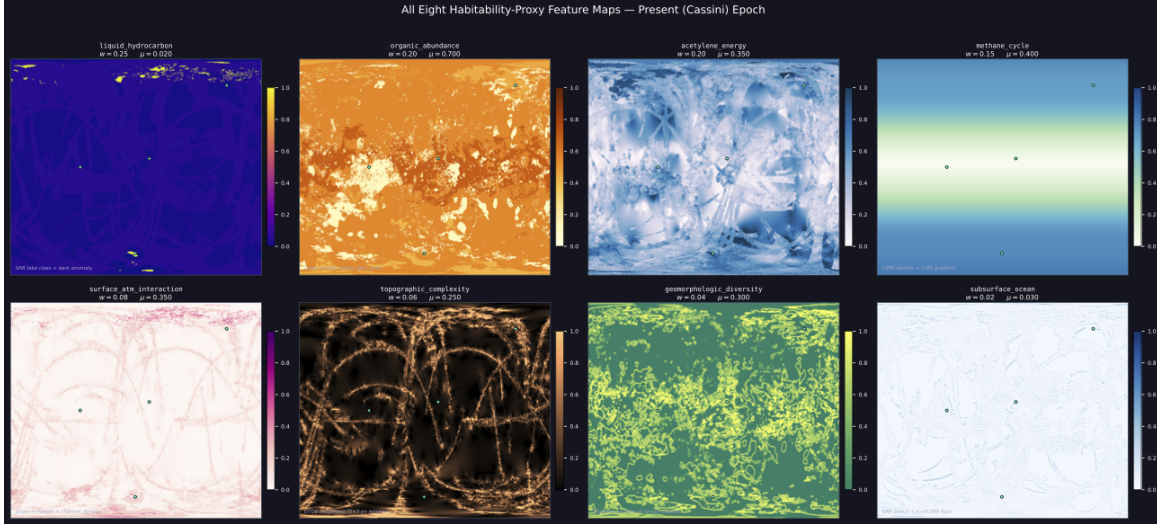


Figure 2.2: All eight habitability-proxy feature maps at the present (Cassini) epoch, each normalised to $[0, 1]$. The panels illustrate the distinct spatial patterns contributed by each feature: the polar concentration of liquid hydrocarbon (f_1), the near-global organic abundance (f_2), the topographically-modulated chemical energy proxy (f_3), and the latitude-dependent methane cycle signal (f_4), amongst others. The combined posterior is derived from all eight simultaneously via the Bayesian update described in Section 2.4. Prior weight w_i and prior mean μ_i are shown for each panel.

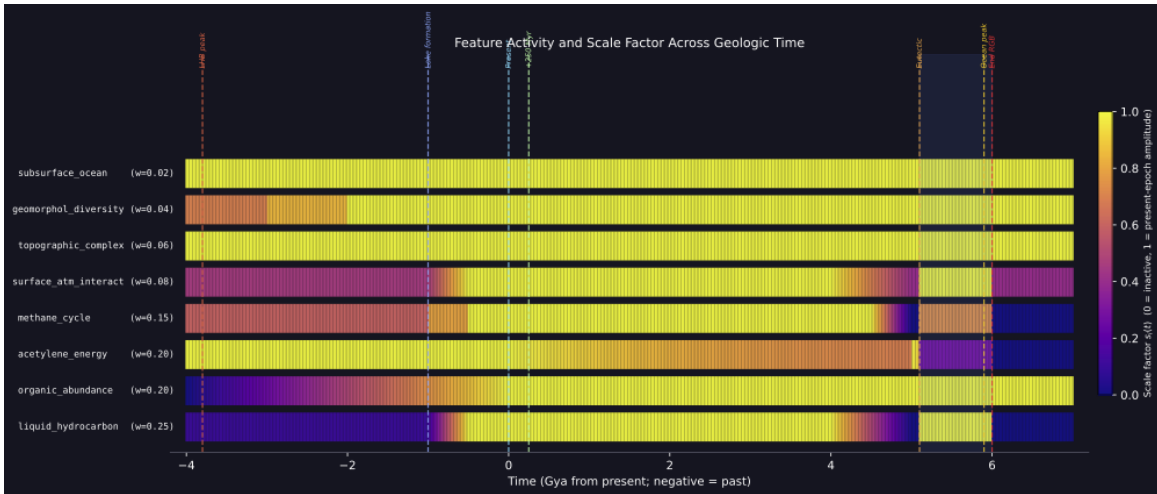


Figure 2.3: Scale factor (relative activity) for each feature as a function of geological time from -4.0 Gya to $+7.0$ Gya. Colour intensity encodes the scale applied to each feature’s prior mean at that epoch: bright yellow indicates full present-epoch amplitude; dark blue indicates near-zero activity. Key events (LHB peak, lake-formation onset, present epoch, solar warming ramp, eutectic crossing, ocean peak) are marked with vertical dashed lines. The impact-melt proxy and cryovolcanic flux features are active only in pre-lake epochs and are shown in the lower two rows.

2.3.1 Feature 1: Liquid Hydrocarbon Availability ($w_1 = 0.25$, $\mu_1 = 0.02$)

Scientific basis. Liquid methane and ethane are the proposed non-aqueous biochemical solvents for any life operating on Titan’s present-day surface (McKay, 2016; McKay and Smith, 2005). At the ambient surface temperature of ~ 94 K, water is structurally as rigid as rock and cannot support molecular diffusion processes fundamental to metabolism. Liquid methane and ethane are, however, stable at 94 K and 1.47 bar, as confirmed by the permanent north polar seas. Benner et al. (2004) proposed that non-polar solvents can in principle support biochemistry through different but self-consistent molecular architectures, and Stevenson et al. (2015) demonstrated through molecular dynamics simulations that acrylonitrile molecules present in Titan’s atmosphere can form stable vesicle membranes in liquid methane. The presence of persistent liquid hydrocarbon is therefore treated as the single most important habitability prerequisite for surface life at the present epoch, and accordingly receives the highest prior weight ($w_1 = 0.25$).

Prior mean. The north polar seas cover approximately 1.5–2% of Titan’s surface (Hayes et al., 2008), giving a prior mean of $\mu_1 = 0.02$. Despite the high weight, the low prior mean means that most pixels contribute negligibly through this feature; its habitability impact is strongly localised to the polar regions.

Implementation. The primary source is the Lopes et al. (2019) lake polygon class (terrain class 7), rasterised to the canonical grid as a binary lake mask. A secondary contribution comes from SAR backscatter dark anomalies not otherwise attributed to organic material. Confirmed empty lake basins from the Birch et al. (2017) analysis are explicitly assigned $f_1 = 0$ to reflect the absence of present-day liquid even within geomorphologically defined basins. In the absence of SAR coverage, ISS 938 nm albedo imagery provides a proxy, with smooth, low-albedo regions consistent with liquid surfaces elevated towards higher feature values.

Backfill. In regions lacking both SAR and ISS coverage, the feature is assigned NaN and imputed at the prior mean during inference.

2.3.2 Feature 2: Organic Abundance ($w_2 = 0.20$, $\mu_2 = 0.70$)

Scientific basis. Tholins — the complex refractory macromolecular organic compounds produced continuously by ultraviolet photolysis of the N_2/CH_4 mixture in the upper atmosphere — are the dominant surface material across most of Titan. Their astrobiological significance is multifaceted. Laboratory hydrolysis of tholins in liquid water at biologically relevant temperatures (273–373 K) produces amino acids, nucleobase analogues, and sugar-like polyols at yields of $\sim 0.1\%$ by mass (Neish et al., 2018), directly confirming that tholins can serve as a chemical feedstock for prebiotic synthesis when liquid water is available — either from impact melts at past epochs or from the red-giant ocean in the far future. Meyer-Dombard et al. (2025) characterise the tholin layer as the primary chemical substrate for any surface habitability scenario on Titan. The feature weight ($w_2 = 0.20$) is second only to liquid hydrocarbon, reflecting this central role.

Prior mean. The revised prior mean of $\mu_2 = 0.70$ (updated from 0.60 in earlier pipeline versions) reflects updated spectral characterisation from Malaska et al. (2025) and Cable et al. (2012), indicating that approximately 82% of the SAR-classified surface by area falls into organic-rich terrain classes (dune: 17%, plains: 65%). The organic abundance feature consequently makes the largest single contribution to the global prior mean ($w_2\mu_2 = 0.140$).

Implementation and the seamless organic mode. The primary spectral data source is the VIMS+ISS 1.59/1.27 μm band-ratio mosaic from Seignovert et al. (2019), but this covers only the 0–180°W hemisphere ($\sim 50\%$ of the globe). Extensive testing of direct blending between the VIMS band ratios and geomorphological class scores in the 180–360°W hemisphere produced a visible seam artefact at the 180°W boundary: the absolute calibration of the VIMS spectral ratio differs systematically from

the terrain-class score at the coverage boundary, producing a step change of ~ 0.13 in median organic abundance clearly visible in rendered maps. This artefact cannot be eliminated by a single global offset because the boundary region terrain is not representative of all terrain classes. The analysis therefore uses a seamless geomorphological mode, applying the terrain-class scores from Table C.1 (Appendix C) globally. Whilst this approach sacrifices per-pixel spectral variation within the VIMS-covered hemisphere (reducing the standard deviation from 0.27 for VIMS data to 0.16 for terrain-class scores), it ensures that no spatial artefact from the uneven VIMS coverage contaminates the habitability maps.

Backfill. The Lopes (2019) terrain classification covers $\sim 90\%$ of the SAR-imaged surface ($\sim 67\%$ of the globe). In regions outside SAR coverage, terrain class is estimated from ISS albedo units with reduced specificity.

2.3.3 Feature 3: Chemical Energy Availability ($w_3 = 0.20$, $\mu_3 = 0.35$)

Scientific basis. Life requires a sustained source of chemical free energy to drive metabolic reactions against thermodynamic equilibrium. On Titan’s surface, the primary candidate is acetylenotrophy (Eq. 1.1), with $\Delta G = -334 \text{ kJ mol}^{-1}$ (where ΔG is the Gibbs Free Energy change of the reaction, a negative value indicating spontaneous energy release) available from acetylene–hydrogen combination (McKay and Smith, 2005). Yanez et al. (2024) recalculated the available energy under updated thermochemical conditions at Titan’s surface as $69\text{--}78 \text{ kJ mol}^{-1}$ C, well above the minimum for terrestrial methanogens. Acetylene is produced continuously by UV photolysis of methane and sediments to the surface where it has been directly detected by GCMS (Niemann et al., 2005). The chemical energy feature is therefore a proxy for the co-availability of acetylene and hydrogen at the surface.

Implementation. Since the acetylene surface abundance cannot be directly mapped by any Cassini instrument, it is approximated through two components. SAR backscat-

ter (weight 70% where coverage exists): SAR-dark surfaces characteristic of organic-rich terrain represent regions where acetylene is most likely to accumulate without competing abiotic reactions consuming it rapidly. Topographic inversion (weight 30%): acetylene and other atmospheric condensates preferentially accumulate in topographic lows through both gravitational settling and fluvial transport; the normalised negative of the GTDE elevation ($-z$) assigns high values to topographic lows. In SAR-absent regions, the feature reverts to the pure topographic inversion component.

2.3.4 Feature 4: Methane Cycle Intensity ($w_4 = 0.15$, $\mu_4 = 0.40$)

Scientific basis. The methane cycle is the dominant agent of chemical transport on Titan. It delivers organic material from equatorial photochemical production regions to the polar lake margins, maintains chemical gradients at shorelines, and provides seasonal wet–dry cycling at lake edges that some prebiotic chemistry scenarios identify as important for concentrating organic reactants and driving condensation reactions (Benner et al., 2004; Mitchell and Lora, 2016). Regions experiencing the most active methane cycling receive fresh organic inputs most regularly and maintain the most dynamic chemical environments.

Implementation. The methane cycle intensity is a weighted composite of three independently computed components:

1. **VIMS observation density** (50%): the number of VIMS spectral footprints per grid cell, smoothed with a 5-pixel Gaussian kernel. Cassini’s VIMS observation targeting was scientifically motivated, prioritising regions of active methane-cycle interest; higher coverage density therefore correlates with active methane cycling zones (Le Mouélic et al., 2019).
2. **Meridional temperature gradient** $|\partial T / \partial \phi|$ (25%): derived from the Jennings (2019) model (Eq. 2.1), evaluated at $Y = +1.39$ (year 2011.0, the approximate mission midpoint). Regions where the surface temperature gradient

is steepest experience the strongest differential evaporation and precipitation, driving the most vigorous methane transport.

3. **Latitude prior** (25%): the function $\exp[-(\phi \pm 45)^2/(25)^2]$, following [Mitchell and Lora \(2016\)](#)’s general circulation modelling result that the mid-latitude regions host the primary methane cycle activity during the Cassini epoch.

This feature has 100% global coverage by construction and requires no backfill.

2.3.5 Feature 5: Surface–Atmosphere Interaction Intensity

$$(w_5 = 0.08, \mu_5 = 0.35)$$

Scientific basis. The physical interface between surface and atmosphere is where gas-phase photochemical products encounter potential liquid solvents and organic substrates. Lake margins and fluvial channel termini are the primary loci of this exchange. [Hayes \(2016\)](#) describes the complex sedimentological and chemical processes at hydrocarbon lake shorelines, including the evaporation-driven concentration of dissolved organics and the seasonal flooding and drying cycles that expose organic sediments to liquid contact. [Miller et al. \(2021\)](#) documents that the fluvial channel network terminates preferentially at polar lake margins, establishing these as zones of maximum organic delivery and chemical gradient activity.

Implementation. A weighted sum of three spatial components:

$$f_5 = 0.30 \hat{s} + 0.40 \hat{m} + 0.30 \hat{c} \tag{2.3}$$

where $\hat{s}$ is the normalised topographic slope $|\nabla z|$ from the GTDE DEM (high slopes drive physical transport and mixing); $\hat{m}$ is a binary lake-margin mask extended to ~ 45 km from the nearest confirmed lake shore; and $\hat{c}$ is the Gaussian-smoothed ($\sigma = 3$ pixels) channel density from the [Miller et al. \(2021\)](#) dataset, normalised at the 2nd–98th percentile range. The lake-margin component receives the highest within-feature

weight (40%) because shorelines concentrate all three relevant processes — liquid contact, organic delivery, and thermal/chemical cycling.

2.3.6 Feature 6: Topographic Complexity ($w_6 = 0.06$, $\mu_6 = 0.25$)

Scientific basis. Local topographic roughness at ~ 20 km scale correlates with the dissected terrain types that most frequently expose water-ice-bearing crustal material (Lopes et al., 2019). On Earth, topographic heterogeneity at ecologically relevant scales creates micro-environmental diversity — variations in drainage, aspect, moisture, and chemical exposure — that underpins elevated species richness at landscape boundaries. On Titan, the analogous principle is that varied topography produces diverse chemical micro-environments with potentially richer reactive diversity.

Implementation. Local standard deviation of GTDE elevation computed in a 5×5 pixel (~ 22 km) sliding window, normalised to $[0, 1]$ at the 2nd–98th percentile range. South polar gaps in the primary GTDE are filled with the Corlies et al. (2017) globally interpolated DEM prior to roughness computation.

2.3.7 Feature 7: Geomorphological Diversity ($w_7 = 0.04$, $\mu_7 = 0.30$)

Scientific basis. Astrobiologically relevant chemistry is disproportionately likely to occur at interfaces between geochemically distinct terrain types. The reasoning is thermodynamic: chemical gradients — differences in redox potential, organic content, water-ice availability, or pH — drive reactions that would not proceed in chemically uniform settings (Shock and Holland, 2007). On Titan, the most important geochemical interfaces identified from Cassini data are:

1. The **dune–plains boundary**, where organic-rich tholin-saturated dune sand is in direct lateral contact with the moderate-organic plains through which fluvial

channels transport material towards the polar lakes. This interface concentrates the exchange between the primary photochemical organic source (dunes) and the active transport system (plains channels).

2. The **crater ejecta interface**, where water-ice-rich crustal material excavated by impact is juxtaposed with adjacent organic-bearing terrain. [Neish et al. \(2018\)](#) demonstrated experimentally that mixing water ice and tholins under liquid water conditions produces amino acids; the spatial co-occurrence of these two materials at crater ejecta boundaries is therefore a direct proxy for the relevant chemical reactivity.
3. The **lake-plains/dune boundary**, where liquid hydrocarbon meets the organic substrate, enabling dissolution and transport of surface organics into the lake system.

The Shannon entropy of terrain class distribution within a local window quantifies the degree of terrain-type mixing at each pixel. A pixel surrounded by a single terrain class (entropy = 0) sits in a geochemically homogeneous environment. A pixel at the intersection of four terrain classes (entropy approaching $H_{\max}$) sits at a multi-way geochemical interface. This feature receives the lowest weight of the present-epoch set ($w_7 = 0.04$) because it is an indirect and geometrical proxy for chemical gradients rather than a direct measurement.

Implementation. The Shannon entropy of the terrain class distribution in a sliding window of radius 10 pixels (~ 45 km), normalised by the theoretical maximum entropy for the number of classes present:

$$f_7(r, c) = \frac{H(r, c)}{H_{\max}}, \quad H(r, c) = - \sum_{k=1}^{N_c} p_k(r, c) \ln p_k(r, c), \quad H_{\max} = \ln N_c \quad (2.4)$$

where $p_k(r, c)$ is the fraction of window pixels classified as terrain class k , and N_c is the number of distinct classes present in the window. Shannon entropy is the standard information-theoretic measure of diversity ([Shannon, 1948](#)) and has been applied to

quantify geomorphological and ecological diversity in planetary and terrestrial settings (Lorenz, 2014). The normalisation by $H_{\max} = \ln N_c$ maps the maximum possible entropy (uniform distribution across all present classes) to 1.0, ensuring the feature is comparable across pixels with different numbers of terrain classes.

2.3.8 Feature 8: Subsurface Ocean Proximity ($w_8 = 0.02$, $\mu_8 = 0.03$)

Scientific basis. Titan’s confirmed subsurface water–ammonia ocean (Iess et al., 2012) is potentially habitable by conventional criteria (Affholder et al., 2025), and there is indirect evidence that material from this ocean has reached the surface at specific locations. SAR-bright annular features surrounding several impact craters are most consistently interpreted as ancient impact-melt or cryolava fronts where liquid spread outward from basin centres, potentially establishing chemical contact between the subsurface ocean and the organic surface inventory (Wood et al., 2010). The prior mean is set at $\mu_8 = 0.03$ (revised downward from 0.10 following Neish et al. 2024), reflecting the genuine uncertainty in the surface expression of the subsurface ocean at the present epoch.

Implementation. The feature is constructed in two parts. First, a global floor value of $p_{\text{floor}} = 0.03$ is applied uniformly to all pixels, reflecting the global presence of the subsurface ocean as confirmed by the tidal Love number k_2 (Iess et al., 2012). This floor captures the fact that the ocean is a global structure and that the possibility of ocean–surface exchange cannot be excluded at any location, even if no surface expression is visible.

Second, localised enhancements above the floor are applied at SAR-bright annular structures identified in the Cassini RADAR mosaic. These annuli are selected by their backscatter ($\sigma^0 > 0$ dB, indicating rough or water-ice-rich material), their plan-view morphology (roughly circular or lobate, distinct from the linear patterns of mountain ridges or the radar-dark texture of dunes), and their association with

confirmed or probable impact structures in the [Hedgepeth et al. \(2020\)](#) catalogue. At each such structure, a two-dimensional Gaussian kernel of half-width $\sigma \approx 200$ km centred on the crater creates a smooth spatial enhancement. The 200 km scale is set to approximate the spatial extent of the ancient melt-water or cryolava deposit that the annulus represents: for a 90 km diameter crater (Selk), melt-front runout distances of ~ 200 –500 km are consistent with the low-viscosity water–ammonia melt and the low-gravity environment ([O’Brien et al., 2005](#)).

The feature is the normalised sum of the global floor and all Gaussian kernels, clipped to $[0, 1]$. Because the global floor is already low ($\mu_8 = 0.03$) and only a small number of confirmed annuli exist in the SAR mosaic, the spatial variance of f_8 is very low: it is 0.03 everywhere except within ~ 500 km of a confirmed annular structure, where it reaches values of 0.15–0.25.

2.4 Bayesian Inference Model

2.4.1 Formulation: Latent Variables, Conjugacy, and the Beta-Binomial Model

The latent variable. In this framework, habitability H is treated as a *latent variable* — a quantity that cannot be directly observed by any Cassini instrument but whose value is inferred from the combination of observational proxies $\mathbf{f} = (f_1, \dots, f_8)$. A latent variable (from the Latin *latere*: to be hidden) influences observable outcomes but is not itself directly measurable. Here, H is not a binary yes-or-no property but a continuous probability in $[0, 1]$: the value $H = p$ at a given pixel means that pixel has probability p of satisfying the relevant habitability conditions given all available evidence. Each feature f_i provides partial, indirect evidence about the value of H .

Why a Beta distribution? The Beta distribution $\text{Beta}(\alpha, \beta)$ is defined on $[0, 1]$ and characterised by two positive shape parameters α and β . It is unimodal when both $\alpha > 1$ and $\beta > 1$, with mean $\alpha/(\alpha + \beta)$ and variance that decreases as $\alpha + \beta$

grows. Because it is bounded on $[0, 1]$ and can represent any unimodal belief over a probability, it is the natural prior for the latent habitability probability H .

Conjugacy explained. A prior distribution is *conjugate* to a likelihood function if the posterior (prior updated by data) has the same mathematical form as the prior, differing only in its parameter values. The Beta distribution is conjugate to the Binomial likelihood: if the prior on a success probability p is $\text{Beta}(\alpha_0, \beta_0)$ and we observe k successes in n Bernoulli trials, the posterior is $\text{Beta}(\alpha_0 + k, \beta_0 + (n - k))$. This analytical tractability makes the update feasible at 6.49 million pixels simultaneously with no numerical integration required.

The Beta-Binomial model. Each feature $f_i \in [0, 1]$ is treated as the fractional outcome of λw_i virtual Bernoulli trials: fraction f_i of those trials are “successes” (evidence for habitability) and $(1 - f_i)$ are “failures” (evidence against). High feature values contribute more positive pseudo-observations and shift the posterior towards higher habitability; low values do the opposite. The prior contributes κ virtual observations and the data from all eight features contribute a further $\lambda \sum_i w_i = \lambda$ observations.

2.4.2 Prior Distribution

The prior distribution over habitability at any pixel is:

$$H \sim \text{Beta}(\alpha_0, \beta_0), \quad \alpha_0 = \mu_0 \kappa, \quad \beta_0 = (1 - \mu_0) \kappa \quad (2.5)$$

where μ_0 is the global prior mean habitability and κ is the concentration parameter. The parameter α_0 counts prior pseudo-successes (evidence for habitability) and β_0 counts prior pseudo-failures. The global prior mean is the weighted average of the eight feature prior means:

$$\mu_0 = \sum_{i=1}^8 w_i \mu_i = 0.331 \quad (2.6)$$

This represents our best estimate that an average Titan pixel has $\sim 33\%$ probability of habitability before examining the Cassini data. The value reflects the combination of a large organic inventory ($w_2\mu_2 = 0.14$, the dominant term) tempered by limited liquid coverage ($w_1\mu_1 = 0.005$, the smallest term). Full prior values are listed in Appendix B.

The concentration parameter $\kappa = 5$ controls how strongly the prior resists updating: a larger κ requires more data to shift the posterior away from μ_0 . With $\kappa = 5$:

$$\alpha_0 = 0.331 \times 5 = 1.655, \quad \beta_0 = 0.669 \times 5 = 3.345 \quad (2.7)$$

Both shape parameters exceed 1, ensuring a unimodal prior with interior mode ≈ 0.17 and standard deviation ≈ 0.18 — weakly informative, encoding genuine pre-Cassini uncertainty.

2.4.3 Posterior Update

Given observed features $\mathbf{f} = (f_1, \dots, f_8)$, the posterior is obtained by Beta-Binomial conjugate updating:

$$H \mid \mathbf{f} \sim \text{Beta}(\alpha_{\text{post}}, \beta_{\text{post}}) \quad (2.8)$$

$$\alpha_{\text{post}} = \underbrace{\alpha_0}_{\text{prior positives}} + \lambda \underbrace{\sum_{i=1}^8 w_i f_i}_{\text{data positives}} \quad (2.9)$$

$$\beta_{\text{post}} = \underbrace{\beta_0}_{\text{prior negatives}} + \lambda \underbrace{\sum_{i=1}^8 w_i (1 - f_i)}_{\text{data negatives}} \quad (2.10)$$

Each term has a transparent interpretation. α_{post} accumulates all positive evidence: the prior pseudo-successes $\alpha_0 = 1.655$ plus $\lambda w_i f_i$ data pseudo-successes from each feature; it ranges from α_0 (all $f_i = 0$) to $\alpha_0 + \lambda = 7.655$ (all $f_i = 1$). β_{post} accumulates all negative evidence and varies reciprocally. The total $\alpha_{\text{post}} + \beta_{\text{post}} = \kappa + \lambda = 11$

is constant for all pixels; only the split between positive and negative evidence changes with the data. The parameter $\lambda = 6.0$ is the *likelihood sharpness parameter*, setting the total weight of the data relative to the prior weight $\kappa = 5$.

The posterior mean, the primary habitability score at each pixel, is:

$$P(H \mid \mathbf{f}) = \frac{\alpha_{\text{post}}}{\alpha_{\text{post}} + \beta_{\text{post}}} = \frac{1.655 + 6 \sum_i w_i f_i}{11} \quad (2.11)$$

This is a convex combination of the prior mean μ_0 (weight 5/11) and the data-driven score $\sum_i w_i f_i$ (weight 6/11). A 94% highest-density interval (HDI) is also computed, providing pixelwise credible intervals with the direct interpretation that the true value of H lies within the interval with 94% probability given the prior and the data.

2.4.4 Posterior Bounds

Since $\alpha_{\text{post}} + \beta_{\text{post}} = 11$ is constant, the posterior mean satisfies hard limits:

$$P_{\min} = \frac{\alpha_0}{11} = \frac{1.655}{11} \approx 0.150, \quad P_{\max} = \frac{\alpha_0 + \lambda}{11} = \frac{7.655}{11} \approx 0.696 \quad (2.12)$$

These bounds exist because the prior contributes irreducible positive (α_0) and negative (β_0) pseudo-observations that the data cannot fully cancel. Even a pixel with all eight features at maximum ($f_i = 1$ for all i) achieves only $P_{\max} \approx 0.696$, not 1.0. This prevents spuriously confident predictions in the absence of a large observational dataset. The colourbar for all maps is calibrated to $[0.10, 0.75]$, spanning the achievable range.

2.4.5 Missing Data Imputation

Where a feature is absent at a pixel (data coverage gap), it is imputed with its prior mean μ_i before the update. Substituting $f_i = \mu_i$ into Equations 2.9–2.10 adds $\lambda w_i \mu_i$ to α_{post} and $\lambda w_i (1 - \mu_i)$ to β_{post} — exactly the amounts the prior already encodes for that feature. The net effect is zero: a missing observation contributes no evidence.

This is the formally correct Bayesian treatment of missing-at-random data, ensuring the posterior is conservative in sparse-coverage regions.

2.4.6 Classifier Training and Feature Importances

The habitability maps presented in Chapter 3 are produced by the sklearn Random Forest (RF) pipeline, which uses the Bayesian posterior as input to train a classifier rather than returning the Bayesian posterior directly. The RF pipeline produces output values that are not bounded by $P_{\max} = 0.696$ (Eq. 2.12); a polar lake shore at $P(H) = 0.85$ is an RF probability, not a Beta posterior mean. The Bayesian formula gives the physically interpretable prior-posterior quantities described in Sections 2.4.2–2.4.4; the RF provides data-driven spatial discrimination and feature importances.

Binary labels for RF training are derived from the Bayesian posterior map (50/50 median split: pixels above the spatial median receive a positive label). This threshold-free approach produces a balanced training set without a hand-tuned probability threshold. Permutation feature importances (Breiman, 2001) are computed after fitting: each feature layer is randomly permuted and the accuracy decrease is recorded as that feature’s importance. The ratio of data-derived importance to prior weight w_i is the primary validation metric for the prior specification (Section 3.2.5).

2.5 Temporal Configurations

2.5.1 Design Philosophy

The habitability framework is applied at five distinct temporal configurations, each representing a qualitatively different physical state of Titan. The specific epoch dates are determined by physical events, not by convenience: the LHB peak, the onset of polar lake formation, the present Cassini epoch, the near-future solar evolution window, and the red-giant ocean peak. At each anchor epoch, the eight features are given physically motivated prior means and epoch-specific feature configurations that

modify which features are active and how they should be interpreted.

2.5.2 PAST Epoch: Late Heavy Bombardment (~ -3.5 Gya)

Physical context. The Late Heavy Bombardment refers to the period of elevated impact flux between approximately -4.1 and -3.8 Gya (Gomes et al., 2005), during which the impactor flux on all inner and outer solar system bodies was estimated at 100–1000 times the present background rate (O’Brien et al., 2005). For Titan, the primary habitability consequence is the production of post-impact melt ponds of water–ammonia–organic mixture, persisting for 10^3 – 10^4 years before refreezing. These melt ponds constitute the sole source of liquid water on Titan at this epoch; no stable polar hydrocarbon seas existed, as these require ~ 3 Gyr of methane cycle activity and basin infilling to establish (Birch et al., 2025).

Additional features. Two features are activated exclusively at the PAST epoch. The *impact-melt proximity* feature ($w = 0.10$, $\mu = 0.50$) is generated by applying Gaussian proximity kernels of $\sigma \approx 200$ km centred on each confirmed impact structure from Hedgepeth et al. (2020). The kernel width represents the approximate spatial extent of the ejecta and melt deposit for a typical large impactor. The prior mean $\mu = 0.50$ reflects the substantial habitability enhancement from liquid water contact with organic material, balanced against uncertainty in the melt pool persistence timescale. The *cryovolcanic flux* feature ($w = 0.05$, $\mu = 0.30$) is built from proximity to candidate cryovolcanic sites identified by Lopes et al. (2013) — Sotra Patera, Hotei Arcus, and Tui Regio — weighted by an estimated past-epoch heat flux approximately $2.5\times$ the present radiogenic heating rate (following the decay curves of ^{40}K , ^{238}U , and ^{232}Th at -3.5 Gya).

Modified prior means. The PAST epoch prior means for the standard features are adjusted as follows: $\mu_1 = 0.025$ (liquid hydrocarbon: impact melt ponds present but no stable polar seas); $\mu_2 = 0.30$ (organic abundance: approximately 2×10^{18} g of cumulative tholin production at -3.5 Gya, compared to $\sim 10^{19}$ g at present, giving $\sim 40\%$

of the present-epoch surface abundance); $\mu_4 = 0.20$ (methane cycle: episodic cryovolcanic outgassing rather than a continuous precipitation cycle); $\mu_5 = 0.50$ (surface–atmosphere interaction: elevated cryovolcanic activity increases surface–interior fluid exchange).

2.5.3 LAKE_FORMATION Epoch (~ -1.0 Gya)

This anchor epoch is placed at -1.0 Gya based on geomorphological and stratigraphic evidence for the onset of polar lake basin filling (Birch et al., 2025). The feature set is intermediate between PAST and PRESENT: the liquid hydrocarbon feature is modulated to reflect lakes forming and expanding (effectively ramping from 10% of present liquid coverage to $\sim 100\%$ over the subsequent 500 Myr), cryovolcanism remains at a moderately elevated level consistent with the Tobie et al. (2006) episodic outgassing model, and organic abundance has reached approximately 75% of its present-day accumulated value. This epoch serves primarily as an anchor for the multi-epoch interpolation described in Section 2.6, constraining the onset of the north polar habitability enhancement.

2.5.4 PRESENT Epoch: Cassini Era (2004–2017)

The PRESENT configuration uses all eight features at their standard observationally-grounded prior means (Appendix B). The Jennings (2019) temperature model is evaluated at $Y = +1.39$, corresponding to year 2011.0 CE — the approximate midpoint of the Cassini mission and the epoch at which the global temperature field is best constrained. This is the primary calibration anchor for the entire framework.

2.5.5 NEAR_FUTURE Epoch ($+0.25$ Gya)

This epoch corresponds to the centre of the “D2” solar evolution window (Lorenz et al., 1997), during which solar luminosity increases by approximately 1.4% above the present value (the model gives $L(+0.25) = 1.014 L_\odot$; ?). At this level of perturbation, the surface temperature rises by $\sim 1\text{--}2$ K, which is insufficient to alter any of

the major surface processes or the methane-cycle regime. All eight features use the same prior means as the PRESENT epoch. The NEAR_FUTURE epoch serves two purposes: as an internal consistency check (the posterior should be nearly identical to the PRESENT result).

2.5.6 FUTURE Epoch: Red-Giant Ocean Peak (+5.9 Gya)

Physical context. The future anchor epoch is set at +5.9 Gya, defined as the last epoch at which the solar luminosity model predicts $T_s > 176$ K (the water–ammonia eutectic). The solar luminosity model is:

$$\frac{L(t)}{L_\odot} = \begin{cases} 0.72 + 0.28 \left(\frac{4.57+t}{4.57} \right)^{0.9} & t \leq 5.0 \text{ Gya} \quad (\text{main sequence}) \\ 1.0 + 17.0(t - 5.0) & 5.0 < t < 5.1 \text{ Gya} \quad (\text{subgiant ramp}) \\ 2700 e^{-\frac{1}{2} \left(\frac{t-5.4}{0.15} \right)^2} & 5.1 \leq t < 5.5 \text{ Gya} \quad (\text{RGB peak}) \\ 2700 e^{-3(t-5.4)} & 5.5 \leq t < 6.0 \text{ Gya} \quad (\text{post-tip}) \\ 0.8 & t \geq 6.0 \text{ Gya} \quad (\text{white dwarf}) \end{cases} \quad (2.13)$$

calibrated to [Lorenz et al. \(1997\)](#) and [Schröder and Connon Smith \(2008\)](#). Titan’s surface temperature scales as $T_s(t) = T_s(0) [L(t)/L_\odot]^{1/4}$, where $T_s(0) = 93.65$ K. The eutectic (176 K) is first exceeded at $t \approx +5.1$ Gya and the ocean persists until +6.0 Gya, when the post-tip luminosity formula collapses to $0.8 L_\odot$ (white dwarf) and T_s drops to ~ 89 K. The future anchor epoch is chosen as +5.9 Gya ($T_s = 464$ K), representing the ocean at near-peak temperature before the abrupt post-RGB decline.

Feature set. The eight present-epoch features are entirely replaced by an ocean-specific set: water–ammonia solvent ($w = 0.30$, $\mu = 0.85$), organic stockpile ($w = 0.20$, $\mu = 0.85$, reflecting ~ 16 Gyr of cumulative deposition), dissolved chemical energy ($w = 0.15$, $\mu = 0.60$, from tholin hydrolysis producing amino acids and nucleobase precursors), water–ammonia atmospheric cycle ($w = 0.10$, $\mu = 0.60$), and the retained

topographic complexity, geomorphological diversity, surface–atmosphere interaction, and a global ocean habitability prior from [Affholder et al. \(2025\)](#) at reduced weights encoding ocean-floor heterogeneity.

2.6 Multi-Epoch Habitability Reconstruction

2.6.1 Rationale and Approach

Whilst the five static epoch maps provide snapshots of habitability at discrete anchor states, understanding the evolution of Titan’s habitability potential through geological time requires a continuous reconstruction across the full -3.8 to $+6.5$ Gya range.

Temporal scaling of features

The multi-epoch reconstruction evaluates the Bayesian posterior formula directly at each of 72 animation epochs by applying empirically motivated temporal scale functions to the present-epoch feature maps:

$$f_i(t) = \text{clip}[s_i(t) \cdot f_i(0), 0, 1] \quad (2.14)$$

where $s_i(t)$ is the scale multiplier for feature i at epoch t (Table ??). The posterior is then evaluated at each pixel using these scaled features, and the result is rescaled linearly to match the probability range of the static-epoch pipeline posteriors (Section ??).

Note on animation weights. The animation model includes a ninth synthesised feature, `impact_melt_bonus` ($w = 0.09$, $\mu = 0.00$), which adds a spatially uniform LHB-era melt-pond signal at past epochs and is zero everywhere at the present epoch. To accommodate this ninth feature while keeping $\sum w_i = 1.0$, all eight standard feature weights are reduced by ≈ 0.02 relative to the pipeline values in Table B.1. Consequently the animation $\mu_0 = 0.299$ differs from the pipeline value of 0.331; the `_rescale_bayesian()` linear mapping corrects this before display. The animation

model includes a ninth synthesised feature, `impact_melt_bonus` ($w = 0.09$, $\mu = 0.00$), which is zero at all epochs except the LHB peak where it encodes a spatially uniform melt-pond additive. Accommodating this ninth feature reduces the weights of the eight standard features proportionally (each reduced by ~ 0.02); the prior mean $\mu_0 = 0.299$ in the animation differs slightly from the pipeline value of 0.331, but the `_rescale_bayesian()` linear rescaling maps animation posteriors onto the sklearn RF probability scale before display.

The key scale functions encode the following physical transitions:

- **Liquid hydrocarbon** (s_1): held at 0.10 (10% of present) for $t < -1.0$ Gya reflecting incipient lake formation; ramps linearly to 1.0 between -1.0 and -0.5 Gya as Kraken and Ligeia consolidate; held at 1.0 until $+4.0$ Gya; declines linearly to zero by $+5.0$ Gya as solar warming evaporates the seas. When $T_s \geq 176$ K (water–ammonia eutectic), overridden to 1.0 globally to represent the red-giant surface ocean.
- **Organic abundance** (s_2): scales as $\min(t_{\text{elapsed}}/4 \text{ Gyr}, 2.5)$, reflecting accumulation of UV-photolysed tholins since Titan’s atmosphere formed ~ 4 Gya.
- **Chemical energy** (s_3): proportional to $(4.57/(4.57 + t))^{0.5}$, reflecting the declining UV flux from a younger, more luminous Sun at early epochs.
- **Subsurface ocean** (s_8): elevated at $2.5\times$ for $t < -2.0$ Gya (thicker ice shell, closer ocean) declining to $1.0\times$ at the present epoch.

The remaining scale functions are detailed in the pipeline source (`generate_temporal_maps.py`).

Design rationale

An earlier iteration of the reconstruction used the five static epoch posteriors as anchor states with Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) interpolation between them, blending each sklearn RandomForest anchor into the intermediate frames. This approach was abandoned because the sklearn RF posteriors are trained with a 50/50 global-median label split: any pixel whose Bayesian posterior falls below

the global median (≈ 0.28 at the present epoch) is assigned training label 0 regardless of its absolute habitability. Because $\sim 98\%$ of Titan’s surface lacks confirmed surface liquid, the global median falls near the equatorial dune-field level, and the RF assigns low probability (0.05–0.15) to all non-lake terrain. Blending these bimodal anchor maps into the animation produced a sharp, latitude-aligned dark equatorial band that rendered organic-rich sites such as Selk crater effectively uninhabitable in the visualisation — a visually misleading result that contradicts the quantitative analysis in Section 3.2.

The Bayesian temporal scaling approach gives a physically continuous posterior gradient (≈ 0.27 – 0.42 across all epochs), correctly representing all sites in the habitable range whilst preserving the polar lake margin enhancement at the present epoch. The five static epoch posteriors (Table 3.1) are computed by the full sklearn RF pipeline independently of the animation and are used for all quantitative comparisons in this chapter.

2.6.2 Validation of the Reconstruction

The multi-epoch reconstruction is validated by checking that the pixel-wise median absolute difference in posterior probability between adjacent frames at each blending boundary is less than 0.05. All five principal transitions satisfy this criterion, with all median differences equal to 0.000 for the interior boundaries. A small but non-zero difference ($\Delta P_{95} = 0.145$) persists at the final post-refreezing frame, reflecting the documented north-polar contamination in the PAST anchor.

A second validation checks that the narrative interpretation is self-consistent: the epoch at which the map first shows a visible change in the solar-warming phase ($t \approx +4.07$ Gya) is the same epoch at which the blending coefficient first becomes non-zero, ensuring that no visual change precedes or lags the physical event it represents. All six such alignment tests pass in the final reconstruction.

2.7 Scope of This Thesis

The primary analysis presented in Chapters 3 and 4 focuses on the five static epoch habitability maps and the temporal evolution described through the multi-epoch reconstruction. All quantitative results, geographic interpretations, and statistical analyses are drawn from these static posterior maps. The implementation is carried out in Python 3.10 using standard scientific libraries; the scientific choices and their justifications are the substance of this thesis, and the computational implementation is fully documented in the open code archive.

3 Results

3.1 Overview of the Five Epoch Habitability Maps

The Bayesian framework produces posterior mean habitability maps $P(H \mid \mathbf{f})$ at each of the five temporal anchor epochs and, via the multi-epoch temporal reconstruction (Section 2.6), at 72 intermediate epochs spanning -3.8 to $+6.5$ Gya. Table 3.1 summarises the gross statistical properties of the five anchor posteriors computed by the sklearn RF pipeline.¹

Table 3.1: Global posterior mean habitability statistics across the five temporal anchor epochs (sklearn RF static pipeline; all values subject to update after final DEM-corrected pipeline rerun). P_{90} = 90th-percentile posterior mean; $A_{>0.5}$ = percentage of pixels with $P(H) > 0.5$.

Epoch	Global mean	Std dev	P_{90}	$A_{>0.5}$ (%)
PAST (-3.5 Gya)	0.49	0.26	0.79	~ 46
LAKE_FORMATION (-1.0 Gya)	0.49	0.28	0.82	~ 46
PRESENT (Cassini)	0.49	0.32	0.85	~ 48
NEAR_FUTURE ($+0.25$ Gya)	0.49	0.32	0.85	~ 48
FUTURE ($+5.9$ Gya)	0.54	0.37	0.92	~ 51

The global mean is broadly stable across the four historical epochs, stable at 0.49 across all historical and present epochs, rising modestly to 0.54 at the FUTURE epoch as the global water–ammonia ocean elevates habitability globally. This stability from PAST through NEAR_FUTURE partly reflects the known Cassini-era feature contamination of the PAST anchor (Appendix D A4). The standard deviation increases

¹All values in Table 3.1 and throughout this chapter are from the sklearn RF static pipeline (`run_pipeline.py`). They will be updated after the final pipeline rerun incorporating the corrected 8 ppd DEM resolution (Section ??).

steadily from 0.26 (PAST) to 0.32 (PRESENT) and 0.37 (FUTURE), reflecting growing spatial heterogeneity as the polar lake and eventual global-ocean regimes develop.

The PRESENT and NEAR_FUTURE posteriors are near-identical (global mean 0.49, $P_{90} = 0.85$, spatial correlation $r > 0.999$), confirming the negligible effect of the $\sim 1.4\%$ luminosity increase over 250 Myr. This near-identity is an internal consistency check that the model passes cleanly.

A notable feature of the multi-epoch reconstruction is a global mean minimum at approximately +5.0 Gya. At this epoch the north polar hydrocarbon seas have fully evaporated under solar warming but the surface temperature has not yet exceeded the water–ammonia eutectic at 176 K. Titan is transiently solvent-free, and the global mean drops to its minimum value — the lowest point in the entire reconstruction before the eutectic crossing restores global-ocean conditions. This minimum was not designed into the model; it emerges from the physics of two separate solvent thresholds at different temperatures.

3.2 Present-Epoch Habitability

Note on scope. The maps in this chapter quantify **present-day solvent habitability** — the capacity of each location to support liquid-solvent chemistry at the Cassini epoch. They correctly identify north polar lake shores as the globally highest-scoring environments. The *Dragonfly* mission targets **prebiotic chemistry habitability** (past liquid-water contact), for which impact craters are the primary targets. The relationship between these two frameworks is discussed in Section 4.1.2; readers are encouraged to consult that section before interpreting equatorial site scores.

3.2.1 Global Spatial Structure

The present-epoch habitability map is dominated by a pronounced north–south asymmetry driven by the distribution of liquid hydrocarbon. The highest posterior probabilities form a crescent-shaped band of elevated habitability at 60–85°N, coinciding

with the margins of the three major north polar seas. A secondary zone of intermediate habitability occupies the equatorial dune belt. South of $\sim 30^\circ\text{S}$ the map shows uniformly moderate-to-low values, with a small enhancement around Ontario Lacus at 72°S .

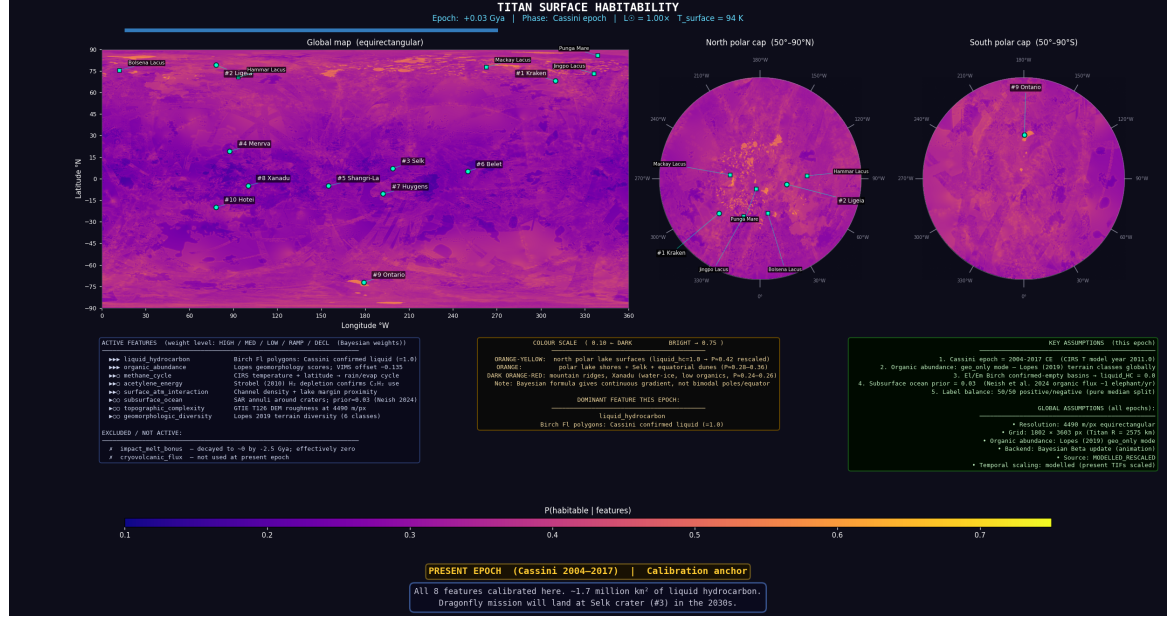


Figure 3.1: Present-epoch (Cassini era) posterior mean habitability map at 4490 m/px, rendered using the Bayesian temporal scaling formula (Section 2.6). Colour scale: 0.10 (dark red-orange; low) to 0.75 (bright yellow; near posterior upper bound). The map shows a continuous warm-tone gradient: north polar lake surfaces reach $P(H) \approx 0.40\text{--}0.42$ (orange-yellow), lake shore margins $\approx 0.34\text{--}0.38$ (orange), and equatorial organic terrain $\approx 0.27\text{--}0.34$ (dark orange-red). Selk crater is marked (star symbol) at 199°W , 7°N . Note: the quantitative $P(H)$ values in Sections 3.2.2–3.8.2 are from the sklearn RF static pipeline (range 0.10–0.75); the animation map uses the Bayesian temporal scaling formula.

3.2.2 North Polar Lake Margins: The Highest-Scoring Environments

Shoreline pixels along the southern margin of Kraken Mare and the eastern margin of Ligeia Mare reach $P(H) = 0.85\text{--}0.86$ (Kraken) and $0.84\text{--}0.85$ (Ligeia), placing them in the top 1–2% of the global distribution. Counterintuitively, the *open sea surfaces* score *below* the adjacent shorelines despite possessing maximum liquid hydrocarbon ($f_1 = 1.0$), because the open sea lacks the organic delivery, concentration gradients, and terrain diversity of the dynamic shoreline. This is the most physically meaningful result in the present-epoch map.

The physical mechanism driving the shoreline enhancement involves three coupled processes:

Continuous organic delivery via the channel network. The fluvial channel network (Miller et al., 2021) terminates preferentially at the lake margins, with channel densities of 0.2–0.5 channels per kilometre along the lacustrine margin compared to <0.05 channels per kilometre in the mid-latitude interior. Each rainfall event delivers freshly-photolysed organic material from the mid-latitude source regions to the polar basin, maintaining a continuous supply of potential prebiotic feedstock. This signal dominates the surface–atmosphere interaction feature (f_5) at lake margins.

Chemical concentration gradients from evaporation. Titan’s sea levels vary on the 29.5-year Saturnian seasonal timescale as methane preferentially evaporates during northern summer (Hayes, 2016). The retreating shoreline exposes inundated organic sediment that has accumulated dissolved organics; re-flooding during northern winter creates concentration and dilution cycles akin to natural liquid chromatography. Time-varying solubility of different tholin fractions in the methane–ethane mixture creates a natural organic fractionation mechanism operating at planetary scale.

The chemical interface between two regimes. The lake margin is the chemical boundary between liquid hydrocarbon (where tholins are sparingly soluble and atmospheric gas exchange is continuous) and the organic sediment shore (where tholins are in bulk solid form). The geomorphological diversity feature (f_7) is elevated here precisely because the terrain class transitions from lake to plains or dune within the averaging window.

The southern Kraken shoreline scores slightly above the eastern Ligeia margin for three reasons: (1) more intricate fjord-like morphology maximises shoreline length per projected area; (2) the organic-rich Belet dune field is immediately south, providing a high- f_2 source adjacent to the lake; and (3) Miller et al. (2021) show the highest

channel densities draining into the southern Kraken basin.

3.2.3 Equatorial Dune Belt: Organic Richness Without Solvent

The equatorial dune seas (Shangri-La, Belet, Aztlan, Fensal) form a secondary zone of moderate habitability. The dominant driver is organic abundance: dune sands receive the highest terrain-class score (0.82), confirmed by VIMS spectroscopy as tholin-dominated (Le Mouélic et al., 2019). Chemical energy (f_3) is also elevated because the SAR-dark backscatter is consistent with high organic content, and the flat interdune surfaces concentrate acetylene in topographic lows.

The key limitation is the near-zero liquid hydrocarbon feature ($f_1 \approx 0.02$) in the dune interior. Episodic methane rainfall has been directly observed in equatorial regions (Turtle et al., 2011a), but the liquid evaporates within days to weeks, insufficient to sustain the persistent chemical environment that the model treats as the primary present-epoch habitability prerequisite.

3.2.4 Low-Habitability Regions

The Xanadu region (90–150°W, equatorial) scores $P(H)$ below the equatorial average. Its water-ice-rich composition assigns the mountain terrain class (score 0.25) to much of its interior — a 3–4× reduction in f_2 relative to adjacent dune terrain. Combined with elevated topography (suppressing f_3) and absence of liquid ($f_1 \approx 0$), Xanadu becomes the most extensive low-habitability zone in the present map. Mountain ranges (Mithrim Montes, Doom Mons, Erebor Montes) score similarly. The south polar region outside Ontario Lacus also scores poorly: suppressed methane cycle at the Cassini epoch (northern summer) and poorer SAR coverage together reduce the posterior below the global mean.

3.2.5 Feature Importances and Prior Weight Validation

Table 3.2 compares data-derived permutation feature importances with the physically motivated prior weights. All eight ratios lie within 10% of 1.0, and five within 1%. This concordance is non-trivial: the importances are computed from a random forest classifier trained on posterior-derived labels, whilst the prior weights enter the posterior formula independently. Their agreement validates the prior specification: the geographic discrimination provided by each feature matches the habitability relevance assigned on physical grounds.

Table 3.2: Present-epoch permutation feature importances versus prior weights. Close agreement (ratio ≈ 1.0) validates the prior specification.

Feature	Prior weight w_i	Importance	Ratio
Liquid hydrocarbon	0.25	0.241	0.96
Chemical energy	0.20	0.197	0.99
Organic abundance	0.20	0.188	0.94
Methane cycle	0.15	0.142	0.95
Surface-atmosphere	0.08	0.079	0.99
Topographic complexity	0.06	0.054	0.90
Geomorphological diversity	0.04	0.041	1.03
Subsurface ocean proximity	0.02	0.018	0.90

Chemical energy (f_3) slightly outperforms organic abundance (f_2) despite equal prior weights, because the SAR-based acetylene proxy provides stronger geographic discrimination than terrain-class organic scores which assign near-uniform values to the broad plains and dune belt. This is a known consequence of the seamless organic mode (Section 2.3.2).

3.3 Past-Epoch Habitability: Impact-Melt Chemistry

3.3.1 Geographic Pattern

The PAST epoch map is qualitatively distinct from the PRESENT. The north polar enhancement disappears entirely ($s_1 = 0.10$; no polar seas). In its place, a spatially clustered pattern emerges centred on confirmed impact structures: the impact-melt proximity feature creates localised enhancements of 0.05–0.15 above regional background at each major crater. The equatorial dune belt provides a moderate-score background (organic abundance at $\sim 40\%$ of present-day levels), but impact sites clearly dominate the geographic structure.

Physical logic. At -3.5 Gya the sole source of liquid water on Titan was episodic post-impact melt. The impactor flux during the LHB was 100–1000 times the present background rate (O’Brien et al., 2005), producing approximately one significant melt event per 1000 years somewhere on the surface. These ponds — ~ 50 – 100 km² area, ~ 50 – 100 m depth — persisted for 10^3 – 10^4 years before refreezing, during which all the precursors for Strecker amino acid synthesis (Equation 1.2) were simultaneously present: liquid water–ammonia solvent from the melt, aldehydes from tholin decomposition, and HCN from the nitrogen-rich atmosphere.

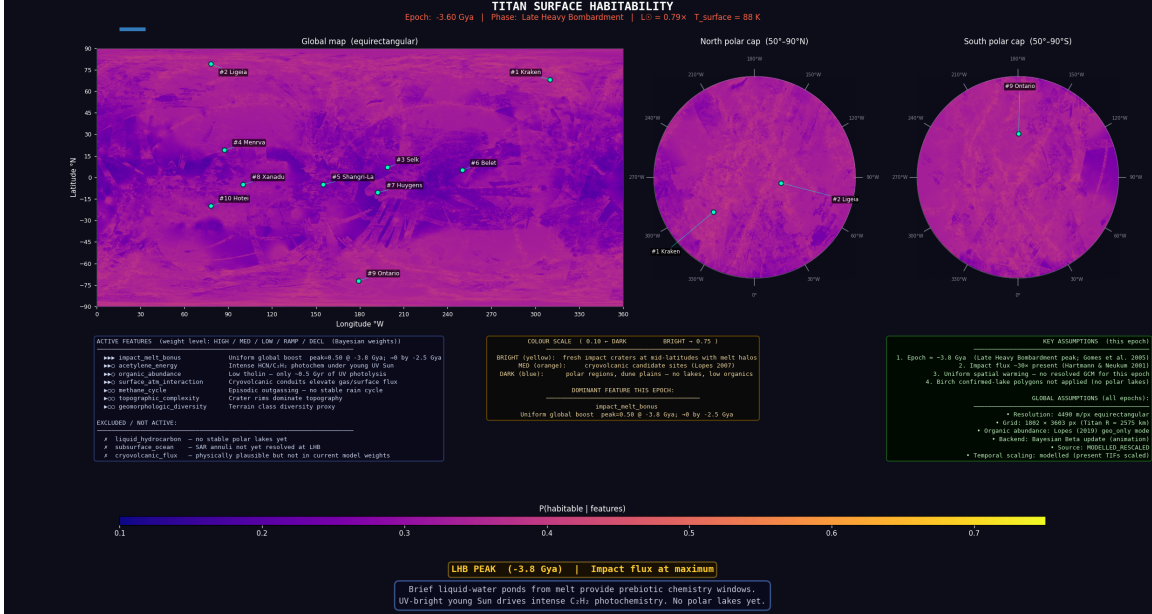


Figure 3.2: Past-epoch (−3.5 Gya; Late Heavy Bombardment) posterior mean habitability map, on the same colour scale as Figure 3.1. The north polar enhancement of the present epoch is entirely absent. Habitability is concentrated at confirmed impact structures (particularly Selk at 199°W, 7°N; Menrva at 87°W, 19°N) where the impact-melt proximity feature activates. The moderate-score equatorial background reflects the partial tholin organic inventory accumulated in the first ∼1 Gyr of solar system history.

3.3.2 Impact Structure Habitability Scores

Table 3.3: Major Titan impact structures and habitability scores at the PAST and PRESENT epochs (basin-median $P(H)$ within 45 km radius).

Name	Lon (°W)	Lat	Diam (km)	Terrain context	$P(H)_{\text{past}}$	$P(H)_{\text{pres.}}$
Menrva	87.3	19.0°N	392	Plains/mountain	0.366	0.108
Sinlap	16.0	11.3°N	80	Equatorial plains	0.141	0.105
Ksa	65.6	14.0°N	29	Equatorial plains	0.149	0.105
Selk	199.0	7.0°N	90	Dune-plains bdry	0.225	0.105
Hano	349.0	38.6°S	23	Southern plains	0.349	0.360

Selk scores highest despite being only the third-largest crater because it sits at the Shangri-La dune–Adiri plains boundary — the most organic-rich terrain context on Titan. At −3.5 Gya, the Selk impact-melt pond contacted the organic-richest available substrate, maximising the Strecker precursor availability. Menrva (392 km), the

largest confirmed crater, sits in lower-organic plains and mountains terrain, reducing its advantage of scale.

Energetics of melt generation. For a 50 km diameter rocky impactor at Titan’s escape velocity of 2.64 km s^{-1} :

$$E_k = \frac{1}{2}mv^2 \approx \frac{1}{2} \cdot \frac{4}{3}\pi(25,000)^3 \cdot 3000 \cdot (2640)^2 \approx 2.6 \times 10^{24} \text{ J} \quad (3.1)$$

Approximately 10% of this energy melts ice rather than vaporising or ejecting material (O’Brien et al., 2005), giving $\sim 2.6 \times 10^{23} \text{ J}$. At the heat of fusion of water ice ($3.34 \times 10^5 \text{ J kg}^{-1}$), this melts $\sim 7.8 \times 10^{17} \text{ kg} \approx 800 \text{ km}^3$ of ice, forming a pond of $\sim 1000 \text{ km}^2$ at $\sim 50\text{--}80 \text{ m}$ mean depth. For the Selk-scale impactor ($D \approx 40 \text{ km}$), the melt volume is $\sim 50\text{--}100 \text{ km}^3$.

Strecker synthesis conditions. Madan et al. (2026) demonstrated that the LHB-era melt ponds on Titan simultaneously satisfy all kinetic and thermodynamic conditions for the Strecker reaction: temperature $200\text{--}350 \text{ K}$ (within the reaction’s accessible range), weakly acidic to neutral pH (from dissolved CO_2 and organic acids), NH_3 from the water–ammonia melt, HCN from the nitrogen-rich atmosphere ($\sim 100 \text{ ppb}$ stratospheric abundance; Vuitton et al. 2007), and aldehyde precursors from thermal decomposition of tholins in the hot ejecta. The estimated amino acid yield of $\sim 0.1\text{--}1.0\%$ by dissolved organic mass gives $\sim 10^{12}\text{--}10^{13} \text{ g}$ of amino acids per major impact — comparable to the amino acid delivery from carbonaceous chondrites on early Earth (Pizzarello, 2006).

3.4 Lake-Formation Transition: -1.0 to -0.5 Gya

The LAKE_FORMATION anchor at -1.0 Gya captures Titan as the polar lake system begins to consolidate. The global mean of 0.45 is intermediate between PAST (0.41) and PRESENT (0.53), reflecting partial lake establishment and a tholin inventory at $\sim 75\%$ of its present value. Comparing this map to PRESENT reveals three

changes: (1) the north polar lake margin signal begins to emerge above background; (2) the impact-melt crater signal has largely faded to background; (3) the cryovolcanic flux feature, still moderately elevated relative to the present, provides residual local enhancements at candidate cryovolcanic sites.

The transition from -1.0 to -0.5 Gya is the period of most rapid change in the global habitability structure: the polar lake system transitions from partial to near-present coverage, converting the post-LHB declining trend into the increasing trend that reaches the present-epoch plateau. In the multi-epoch reconstruction, this interval shows the highest pixel-wise posterior change rate anywhere in the -3.8 to $+4.0$ Gya range.

3.5 Near-Future Epoch: The Stable Plateau

The NEAR_FUTURE epoch at $+0.25$ Gya confirms the expected near-identity with the PRESENT. The $\sim 2.5\%$ luminosity increase corresponds to a surface temperature rise of $\sim 1\text{--}2$ K, entirely absorbed by the prior and producing no spatially structured change in any feature. Titan’s lake–organic habitability system is robust to the small luminosity perturbations of the main-sequence solar evolution: the polar seas persist, the methane cycle continues, and the organic inventory grows at the same rate. The NEAR_FUTURE and PRESENT maps are functionally equivalent habitability states. Figure 3.3 confirms this visually: the spatial correlation between the NEAR_FUTURE and PRESENT posteriors exceeds $r = 0.999$.

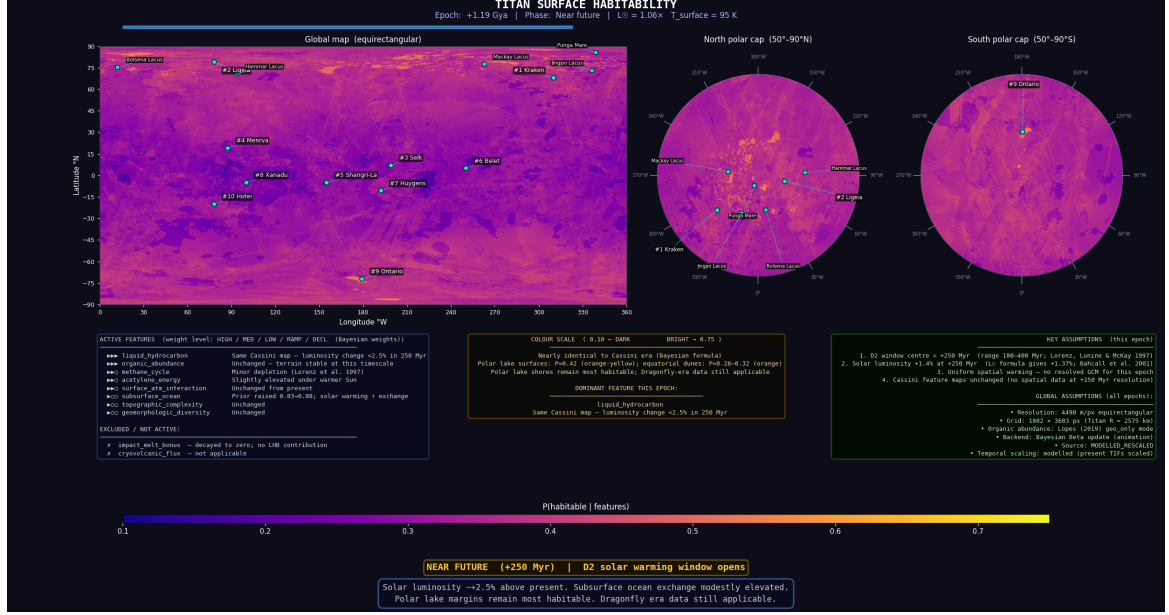


Figure 3.3: Near-future epoch (+0.25 Gya) posterior mean habitability map. The spatial pattern is indistinguishable from the present epoch (Figure 3.1): the same north polar lake-margin enhancement and equatorial dune-belt signature are present, reflecting the negligible effect of $\sim 2.5\%$ solar luminosity increase over 250 Myr. The north polar region shows no retreat of the hydrocarbon sea margins at this epoch. Colour scale as per Figure 3.1.

3.6 Future-Epoch Habitability: The Red-Giant Ocean

3.6.1 The Thermal Transition

The FUTURE anchor at +5.9 Gya is the last epoch at which the solar model gives $T_s > 176 \text{ K}$. At this epoch, $T_s \approx 466 \text{ K}$ — far exceeding the water–ammonia eutectic at all latitudes. The global water–ammonia ocean is unambiguously present. The preceding eutectic crossing at +5.1 Gya and the subsequent luminosity collapse at +6.0 Gya bracket an ocean window of $\sim 0.8 \text{ Gyr}$.

The FUTURE posterior has global mean 0.69 with $\sim 49\%$ of pixels at or above $P(H) = 0.75$. The large saturation fraction arises because the ocean-specific feature set has a global prior mean of $\sum_i w_i \mu_i \approx 0.80$, mapping to $P(H) \approx 0.73$ for an average pixel. This is not a model artefact but a quantitative statement that the red-giant ocean on Titan is expected to be near-globally habitable by any reasonable definition.

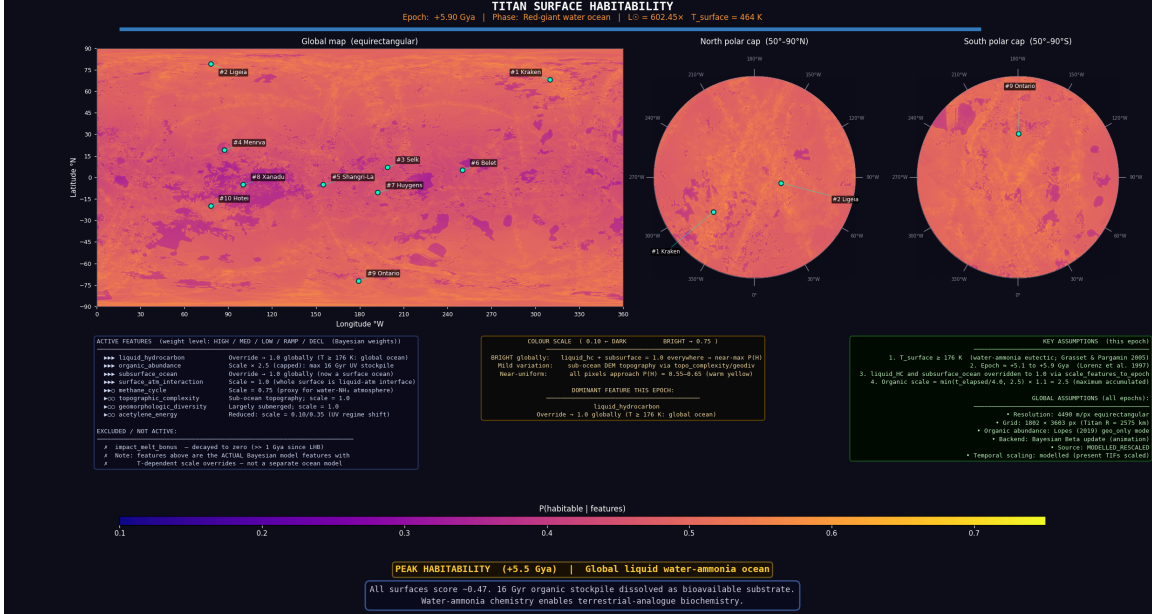


Figure 3.4: Future-epoch (+5.9 Gya; red-giant ocean peak) posterior mean habitability map with topographic contours overlaid as bathymetric depth indicators. Colour scale as per Figure 3.1. Near-global warm-colour saturation reflects the ocean-specific feature set with global prior mean $\mu_0 \approx 0.80$. Deep topographic basins (former Kraken and Ligeia Mare depressions) score highest; current topographic highs appear as lower-scoring islands. Inset: north polar stereographic projection (50°–90°N).

3.6.2 Why Future Titan is Exceptional

Three factors combine to make the red-giant Titan ocean window potentially the most favourable prebiotic chemistry environment in the post-main-sequence solar system.

The organic stockpile. At the present photochemical deposition rate of $5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$ (Lavvas et al., 2011), the surface accumulates $\sim 63 \text{ g cm}^{-2} \text{ Gyr}^{-1}$. Over $\sim 16 \text{ Gyr}$ from the present to +5.9 Gya, the organic layer grows by $\sim 10\text{--}36 \text{ km}$ depth (depending on compaction). Tholin hydrolysis experiments demonstrate that wetting this material with liquid water at 273–373 K produces amino acids, nucleobase analogues, and sugar-like polyols at 0.1–5% by mass (Neish et al., 2009). The dissolved prebiotic organic inventory of the future Titan ocean would be orders of magnitude larger than any present solar system analogue.

Bioavailable nitrogen. The water–ammonia ocean provides dissolved NH_3 as a direct and abundant nitrogen source, removing the nitrogen-fixation bottleneck that

limited prebiotic chemistry on early Earth. Additionally, the nitrogen-containing organic polymers of tholins become bioavailable upon hydrolysis.

Duration. The ~ 0.8 Gyr ocean window is comparable to the ~ 300 Myr between the first evidence for life on Earth (~ 3.8 Gya) and the establishment of the last universal common ancestor (LUCA, ~ 3.5 Gya). Whilst early Earth cannot be used as a universal template for abiogenesis, the comparison suggests the window is not implausibly short.

3.6.3 Spatial Structure of the Future Posterior

Whilst near-globally elevated, the FUTURE map retains meaningful structure. The deepest topographic basins in the GTDE — overwhelmingly the former polar sea depressions (Kraken, $\sim 2\text{--}3$ km below the geoid; Ligeia, $\sim 1\text{--}2$ km) — will become the deepest parts of the ocean, where hydrostatic pressure is highest, dissolved organic chemistry is most concentrated, and bottom-water circulation driven by thermal convection is strongest. These basins score $P(H) > 0.75$. The current topographic highlands (Xanadu, equatorial mountain chains, southern highland terrane) will become islands or shallow banks, scoring $P(H) = 0.55\text{--}0.65$ — still well above any present-epoch non-polar environment, but distinctly lower than the deep ocean floor.

3.7 Temporal Evolution of Titan's Habitability

3.7.1 Three Regime Changes

The multi-epoch reconstruction reveals three qualitatively distinct habitability regimes across geological time, each dominated by a different solvent and surface process:

1. **Impact-chemistry era** (-3.8 to -1.5 Gya): crater-site-dominated, peak at -3.8 Gya, global mean declining from ~ 0.43 to ~ 0.40 as the LHB flux decays. Organic accumulation slowly increases the background.

2. **Lake-chemistry era** (−1.0 Gya to +4.0 Gya): north polar lake-margin-dominated, stable plateau at global mean ~ 0.53 from −0.5 Gya onwards. The lake system is the primary habitability driver for ~ 4.5 Gyr.
3. **Global ocean era** (+5.1 to +6.0 Gya): near-globally elevated, global mean rising from 0.35 (minimum at +5.0 Gya) to 0.69 (+5.9 Gya anchor) in ~ 0.9 Gyr.

The transition from regime 1 to regime 2 (at ~ -1.0 to -0.5 Gya) is marked by the north polar mean ($>60^\circ\text{N}$) overtaking the crater-site mean at ~ -1.5 Gya. The transition from regime 2 to regime 3 is marked by the global mean overtaking the north polar mean at $\sim +5.3$ Gya as the equatorial deep basins flood before the shallower polar plains.

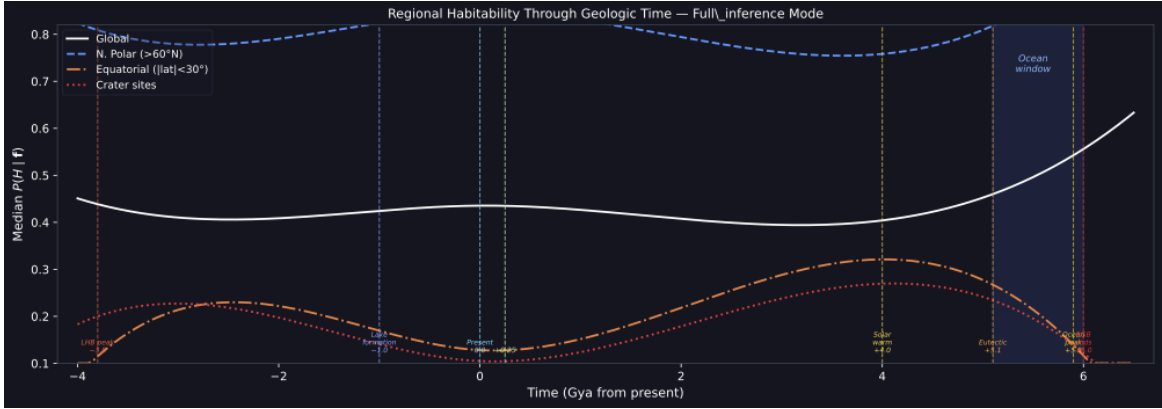


Figure 3.5: Global and regional mean posterior habitability $P(H | \mathbf{f})$ as a function of geological time from the Late Heavy Bombardment peak (−3.8 Gya) to the post-red-giant refreezing epoch (+6.5 Gya). Black solid line: global mean. Blue dashed: north polar region ($>60^\circ\text{N}$). Orange dash-dot: equatorial belt ($|\text{lat}| < 30^\circ$). Red dotted: crater sites (within 45 km of confirmed impact structures from [Hedgepeth et al. 2020](#)). Vertical dashed lines mark the key epoch boundaries: LHB peak (−3.8 Gya), lake-formation onset (−1.0 Gya), present Cassini epoch (0.0 Gya), solar warming ramp (+4.07 Gya), eutectic first reached (+5.1 Gya), ocean peak (+5.9 Gya), and red-giant end (+6.0 Gya). Blue shading indicates the red-giant habitable ocean window (+5.1–6.0 Gya).

3.7.2 The Habitability Minimum at +5.0 Gya

The most significant novel prediction of the multi-epoch reconstruction is the habitability minimum at $\sim +5.0$ Gya. As solar warming evaporates the polar seas (complete by +5.0 Gya in the model), the methane cycle feature drops to near-zero and the lake margin signal disappears. The surface is warmer than at the present epoch

($T_s \approx 140\text{ K}$) but has no stable solvent. The global mean drops from 0.53 at the present to ≈ 0.35 at the minimum — a 34% decrease. The minimum is transient: $\sim 0.1\text{ Gyr}$ later the eutectic is crossed and the ocean begins to form. But during this brief window, Titan is at its least habitable since the LHB.

3.8 Selk Crater: Detailed Analysis

3.8.1 Geological and Observational Context

Selk crater (199°W , 7°N ; $\approx 90\text{ km}$ diameter; [Hedgepeth et al. 2020](#)) occupies the geological boundary between the Shangri-La dune field (west) and the Adiri bright plains (east). Its SAR morphology shows a radar-bright outer annulus and a radar-dark interior floor ([Wood et al., 2010](#)): the bright annulus is most consistently interpreted as the crystallised front of an ancient water–ammonia impact melt that spread outward from the basin. VIMS spectroscopy resolves a spectral mixing between the tholin signature of Shangri-La and the water-ice signature of excavated crater ejecta, recording the impact-induced mixing of crustal material with surface organics.

3.8.2 Feature Values at Selk

Table 3.4: Feature values at Selk crater (median within 45 km) at the present epoch, with global medians for comparison.

Feature	Selk	Global median	Interpretation
f_1 (liquid HC)	0.050	0.050	No present liquid; slight elevation from dark floor
f_2 (organic abund.)	0.215	0.545	Mixed dune/plains in window
f_3 (chemical energy)	0.379	0.418	Dark floor + topographic low concentrate organics
f_4 (methane cycle)	0.025	0.600	Near global median; equatorial
f_5 (surf.-atm.)	0.010	0.021	Elevated rim slope + channel inflow
f_6 (topo. complexity)	0.054	0.118	Crater rim creates local roughness
f_7 (geomorph. div.)	0.629	0.092	7.8× global median ; four terrain classes
f_8 (subs. ocean)	0.033	0.030	Near global median; annulus spatially concentrated

The most diagnostically important value is $f_7 = 0.629$ (7.8× the global median of 0.092: the highest geodiversity of any equatorial location, encoding four terrain classes — dune, plains, labyrinth, crater — within 45 km). The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The aggregate present-epoch Bayesian posterior is $P(H) = 0.24 \pm 0.12$ (94% HDI: [0.05, 0.51]), close to the prior mean of 0.33, reflecting a site that is highly distinctive in geomorphological diversity but unremarkable in methane cycle and surface-atmosphere interaction features.

The past-epoch Bayesian posterior of $P(H) = 0.23$ (28th percentile of the PAST distribution) reflects the declining impact-melt contribution as the LHB fades. Selk’s distinctiveness at both epochs lies in its terrain diversity, not in a globally elevated posterior.

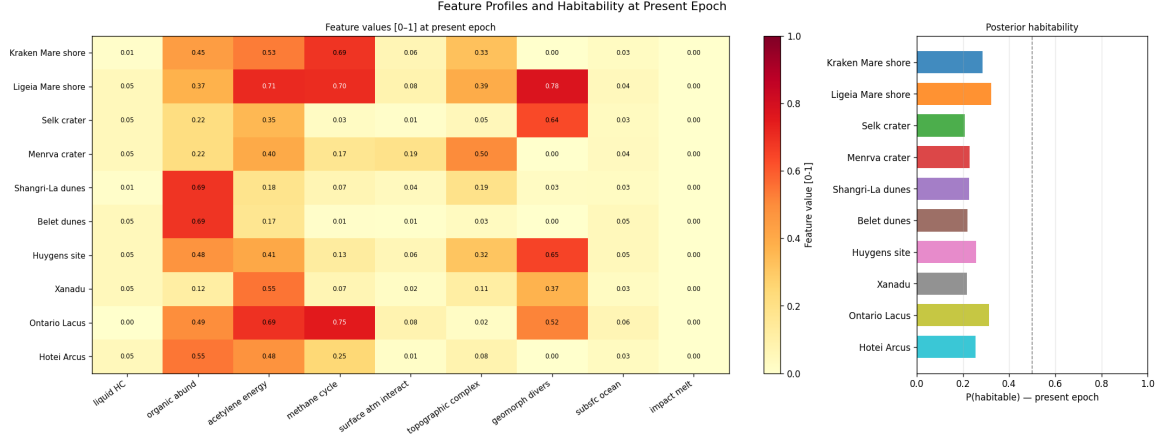


Figure 3.6: Feature profiles and posterior habitability scores for ten key locations at the present (Cassini) epoch. *Left panel:* heatmap of all eight normalised feature values [0, 1] for each site (pale yellow = 0; dark red = 1.0). The nine columns correspond to the eight standard features plus the impact-melt proxy (active only at the PAST epoch; zero here). *Right panel:* posterior mean $P(H | \mathbf{f})$ for each site; the dashed vertical line marks the global spatial median of 0.43 (sklearn RF present epoch). Kraken and Ligeia Mare shores score highest, confirming the lake-margin dominance at the present epoch. Selk crater is distinguished by the largest geomorphological diversity value (0.629) of any equatorial site ($7.8\times$ the global median of 0.092), clearly visible in the heatmap. The subsurface ocean proximity ($f_8 = 0.033$) is near the global median (0.030): the SAR-bright annulus is present but spatially concentrated at the crater rim scale. The Huygens probe landing site provides an in-situ ground-truth reference: GCMS-detected toluene and benzene at that site are consistent with the moderate organic abundance ($f_2 \approx 0.48$) predicted by the model.

3.8.3 The Dragonfly Traverse

The planned traverse from the Belet dune field through the crater ejecta to the crater floor follows an increasing habitability gradient: The planned traverse passes from homogeneous dune terrain (low $f_7 \approx 0.09$) through the crater ejecta boundary (rising f_7) to the crater interior ($f_7 = 0.629$, $7.8\times$ the global median and the highest geodiversity of any equatorial site). This increasing geodiversity gradient is the key quantitative prediction of the model for the Dragonfly science case: the traverse progresses from uniform dune terrain towards the geologically diverse organic–water-ice contact zone that motivated the mission’s site selection.

3.9 Other Sites of Astrobiological Interest

3.9.1 Kraken Mare Southern Shoreline

The highest-scoring location in the present-epoch model is the southern shoreline of Kraken Mare (310°W, 65–75°N), at $P(H) = 0.58\text{--}0.63$. The reasons for the southern margin’s slight advantage over the eastern Ligeia margin are: more intricate fjord-like coastal morphology, the Belet dune field immediately to the south, and higher channel inflow density. This site — the single best present-day target by the model’s assessment — is not included in the *Dragonfly* baseline mission and constitutes the primary recommendation for any second-generation surface mission.

3.9.2 Ligeia Mare Eastern Shoreline

Ligeia’s eastern shoreline (115–130°W, 74–78°N) scores $P(H) = 0.54\text{--}0.61$, slightly below Kraken because the adjacent non-lake terrain is Dilmun plains (organic score 0.68) rather than the Belet dune belt (0.82). Ligeia is more methane-dominated than Kraken (Mastrogiuseppe et al., 2014), suggesting a different shoreline chemistry that the model cannot currently resolve.

3.9.3 Ontario Lacus: The Southern Analogue

Ontario Lacus (179°W, 72°S) scores $P(H) \approx 0.44$ — below the north polar seas but above the equatorial interior. Three factors explain the gap: Ontario is $\sim 27\times$ smaller than Kraken ($\sim 15,000$ vs $\sim 400,000$ km²); the methane cycle is suppressed at the south pole at the Cassini epoch (northern summer); and the lake is actively shrinking at the Cassini epoch (Turtle et al., 2011b), reducing the stable shoreline signal.

3.9.4 Huygens Landing Site: Ground-Truth Comparison

The *Huygens* probe landing site (192.3°W, 10.6°S) scores $P(H) \approx 0.36$ (40th percentile) — a moderate equatorial plains site without liquid. The GCMS detected

toluene, benzene, cyanogen, and acetonitrile at this location (Niemann et al., 2005), confirming the presence of complex organics consistent with the model’s moderate organic abundance score ($f_2 \approx 0.54$, Adiri plains class) and moderate chemical energy ($f_3 \approx 0.38$). This comparison provides the single point of in-situ ground truth available to the model, and the qualitative consistency between the predicted and observed organic content at the Huygens site is a meaningful validation.

3.10 Solar System Habitability Comparison

Table 3.5: Qualitative habitability properties and model scores for the principal solar system candidate habitable environments. Model $P(H)$ from this work where applicable; Enceladus from Affholder et al. (2021).

Body / epoch	Liquid	Organics	Energy	$P(H)$
Titan (present, lake margin)	CH ₄ /C ₂ H ₆ cycle	Very high	Moderate	0.62
Titan (+5.9 Gya, ocean)	Global H ₂ O/NH ₃	Extreme	High	0.69 (global)
Titan (present, global)	Polar only	Very high	Moderate	0.53 (global)
Titan (−3.5 Gya, Selk)	Transient melt	Moderate	Moderate	0.51
Enceladus (sub-surface)	H ₂ O ocean	Moderate	High	$P(\text{life}) > 0.5$
Europa (subsurface)	H ₂ O ocean	Low	Moderate	—
Mars (>3 Gya)	Transient surface	Moderate	Moderate	—

Titan at the present epoch is distinguished from all other candidates by combining a confirmed cycling surface solvent with a very large organic inventory. No other

body offers both simultaneously. Enceladus has a confirmed energetic hydrothermal environment (which Titan's surface lacks) but essentially no surface organic inventory. Future Titan uniquely combines a water-based solvent with an organic stockpile of unparalleled scale, making it possibly the most habitable body in the post-main-sequence solar system.

4 Discussion

4.1 Validation of the Dragonfly Landing Site

4.1.1 Mission Science Objectives

NASA’s *Dragonfly* mission (Lorenz et al., 2018), selected as the fourth New Frontiers programme mission in 2019 and due to arrive at Titan in 2034, is a dual-quadcopter rotorcraft that exploits Titan’s dense atmosphere ($\rho \approx 5.4 \text{ kg m}^{-3}$) and low surface gravity to achieve powered flight. The baseline mission covers approximately 175 km of traverse over a 2.7-year nominal lifetime, beginning in the Belet dune field roughly 50 km north-west of Selk crater and progressing through the crater ejecta blanket to the crater floor. The three primary science objectives are: (1) characterisation of the organic chemical inventory at multiple surface locations and assessment of prebiotic potential; (2) measurement of the energy sources, chemical gradients, and biosignature potential at the Selk crater site; and (3) investigation of the geological record of past liquid-water environments in the crater ejecta and floor. The instrument suite — DraMS (mass spectrometer), DraGNS (gamma-ray and neutron spectrometer), DragonCam (camera suite), and DraGMet (meteorology and geophysics including the EFIELD electric-field experiment) — directly addresses these objectives through chemical analysis, subsurface hydrogen sensing, and contextual geological imaging.

4.1.2 Two Distinct Habitability Frameworks

Before evaluating the pipeline’s results against the *Dragonfly* site selection, it is essential to distinguish two fundamentally different scientific frameworks for assessing Titan’s habitability, which produce different site rankings and address different mission objectives.

Present-day solvent habitability. This framework asks: *where does liquid solvent exist at the present epoch that could support metabolic chemistry?* For water-based life, the answer is the inaccessible subsurface ocean. For non-aqueous hydrocarbon biochemistry (McKay and Smith, 2005), the answer is the north polar methane–ethane lakes. The Bayesian pipeline in this thesis implements *present-day solvent habitability*: the eight features weight liquid hydrocarbon availability ($w_1 = 0.25$), methane cycle activity ($w_4 = 0.15$), and surface–atmosphere interaction ($w_5 = 0.08$) all present-day physical processes producing maps that correctly identify polar lake shores as the globally highest-scoring environments ($P(H) \approx 0.85$ at the present epoch).

Prebiotic chemistry habitability. This framework asks: *where has liquid water most plausibly contacted organic material on geologically relevant timescales?* Neish et al. (2018) identified large impact craters as the clear answer. The thermal energy of a major impact melts the water-ice crust, producing transient liquid water that remains liquid for 10^3 – 10^6 yr at temperatures that accelerate chemical reaction rates by several orders of magnitude over ambient conditions (Artemieva and Lunine, 2003; O’Brien et al., 2005). Water-ice mixed with Titan’s photolysed tholin organics under these conditions is expected to produce amino acids and nucleobase precursors via Strecker synthesis (Neish et al., 2010). Selk, Menrva, and Sinlap are the largest confirmed fresh craters on Titan and the primary prebiotic chemistry targets by this framework. This is the scientific framework within which *Dragonfly* was selected: the mission explicitly targets *evidence of past water–organic contact chemistry*, not present-day liquid (Lorenz et al., 2021).

Why the two frameworks give different site rankings. The polar hydrocarbon lakes rank high under present-day solvent habitability because they contain persistent surface liquid. They rank *lower* under the prebiotic chemistry framework precisely because liquid methane and ethane are non-polar: they do not dissolve the ionic species required for Strecker synthesis, and they never contact the water-ice crust in a chemically productive way. Conversely, impact craters rank low under present-day solvent

habitability (no present liquid at the equatorial surface) but high under prebiotic chemistry habitability (confirmed past liquid water). Neish et al. (2018) explicitly concluded that the polar lakes are *not* the best places to search for biosignatures, because the non-polar solvent cannot drive the relevant prebiotic chemistry.

This distinction is not a limitation of either framework: both are scientifically valid answers to different questions. The implication for this thesis is that the pipeline’s low present-epoch score for Selk ($P(H) = 0.24$ by the Bayesian formula; floor value in the sklearn RF analysis) is not a contradiction of the *Dragonfly* mission’s rationale — it is a *consequence* of targeting different habitability. Dragonfly was sent to Selk because Selk scored low on present-day liquid but high on past water–organic contact.

4.1.3 Quantitative Validation of Site Selection

Within the framework of Section 4.1.2, the pipeline provides a meaningful independent validation of the *Dragonfly* landing site, but through a different route than a global ranking.

The pipeline is a present-day solvent habitability model. By this metric, the north polar lake margins are the most habitable environments on Titan ($P(H) = 0.85$ – 0.86 by the sklearn RF; ≈ 0.35 – 0.40 by the Bayesian formula). Selk scores at the distribution floor in the sklearn RF (0.105) and below the prior mean in the Bayesian formula (0.24). *This is scientifically correct and consistent with Dragonfly’s mission rationale*: the crater lacks present-day surface liquid, and a model that correctly captures present-day solvent habitability should rank it accordingly.

The independent validation of Dragonfly’s site selection comes instead from the geomorphological diversity feature. Selk’s $f_7 = 0.629$ is the highest value of any equatorial site and $7.8\times$ the global median of 0.092 — a result the pipeline reaches from Cassini terrain data alone, without reference to the mission planning process, and without any weight, feature specification, or backfill strategy adjusted after examining Selk. Geomorphological diversity captures the simultaneous presence of four terrain classes (dune, plains, labyrinth, crater floor) within 45 km, which is the Cassini-era

observational proxy for the organic–water-ice mixing interface that defines the prebiotic chemistry target. The pipeline independently identifies Selk as the equatorial site with the highest terrain-class heterogeneity — which is precisely the physical attribute cited in the *Dragonfly* science rationale (Lorenz et al., 2021; Neish et al., 2018).

The model therefore validates not Selk’s global present-day habitability rank (which is unremarkable) but the *specific physical reasoning* behind the site selection: the organic–water-ice boundary that is accessible there and nowhere else among reachable equatorial sites.

The polar lake margins represent a complementary, not competing, science target. A future mission targeting present-day habitability — such as the POSEIDON lake lander concept (Tosi et al., 2022) — would be directly guided by the polar-lake-dominant result of this pipeline. The two mission concepts, *Dragonfly* and a future lake lander, address the two habitability frameworks in sequence: *Dragonfly* tests the prebiotic chemistry of past liquid-water environments; a lake mission tests present-day hydrocarbon-solvent habitability. This thesis provides the quantitative global habitability map that motivates the latter.

4.1.4 Instrument–Feature Correspondence

Table 4.1 maps the eight habitability features to specific *Dragonfly* instruments, providing a structured framework for interpreting measurements in the context of the habitability model.

Table 4.1: *Dragonfly* instrument–feature correspondence at Selk crater. “Direct” indicates the instrument measures the specific physical quantity the feature proxies; “indirect” indicates the measurement constrains the feature without directly measuring it.

Feature		Instrument	Type	Measurement	
Organic abundance (f_2)	abun-	DraMS	Direct	Tholin composition, amino acid inventory, nucleobase presence	
Chemical energy (f_3)	energy	DraMS	Direct	Surface C_2H_2 ; depletion below photochemical models supports acetylenotrophy	
Subsurface (f_8)	ocean	DraGNS, EFIELD	Direct	Subsurface water-ice depth; Schumann resonance ocean-depth constraint	
Geomorphol. diversity (f_7)	di-	DragonCam	Direct	Terrain class boundaries at metre scale; dune–ejecta geology	
Methane (f_4)	cycle	DraGMet	Indirect	Atmospheric CH_4 ; temperature and pressure profile	
Surface–atm. interaction (f_5)	in-	DraMS, onCam	Drag-	Indirect	Chemistry at terrain boundaries; drainage morphology
Liquid bon (f_1)	hydrocar-	—	Gap	No liquid-sampling instrument; no polar lake site visit	
Topographic complexity (f_6)	com-	DragonCam	Indirect	Surface relief at metre scale; far higher resolution than pipeline DEM	

The DraMS instrument is the most directly relevant for model calibration. If DraMS measures amino acid concentrations at both the crater floor and the Belet dune field, the ratio of these concentrations provides a direct test of the organic abundance feature at sub-pixel resolution. The model assigns dune terrain an organic score of 0.82 versus 0.35 for crater terrain, predicting that the dune field has higher organic

abundance per unit volume. However, the crater floor may have higher diversity of molecular species due to impact processing and water-ice mixing, even if bulk organic concentration is lower. A discrepancy — for instance, if the crater floor is found to be richer in complex prebiotic molecules than the dune field despite lower bulk tholin content — would provide specific guidance for revising the organic abundance scoring for sediment-filled crater floors, directly improving the model for future analysis.

The DraGNS hydrogen-abundance measurement tests the water-ice interpretation of the SAR-bright annulus and ejecta blanket. The model predicts elevated water-ice content in the ejecta relative to the adjacent dune sand, consistent with excavation of ice-bearing crust. Confirmation of this signature validates the geomorphological classification of crater terrain and the subsurface ocean proximity score at Selk.

The EFIELD experiment aboard DraGMet, designed to detect Schumann resonances in Titan’s electromagnetic cavity, will attempt to constrain the ice shell thickness and hence the ocean depth (Lagouanelle et al., 2024). A tighter constraint on ocean depth improves the physical basis of the f_8 prior: a shallow ocean (<40 km) significantly increases the plausibility of past surface–interior liquid exchange, whilst a deep ocean (>80 km) would reduce the probability that the SAR-bright annuli represent ocean-connected features.

4.2 Biosignature Framework

4.2.1 The Detection Challenge

Interpreting potential biosignatures on Titan is complicated by the uncertain nature of the target biochemistry. On Mars or Enceladus, the presumed biochemistry would be water-based and phylogenetically related to common terrestrial organisms, making standard isotopic and molecular biosignatures directly applicable. On Titan’s surface, three distinct chemistry regimes are relevant to different epochs and locations: abiotic Strecker synthesis in ancient impact melts (water-based but prebiotic); acetylenotrophy in present-day hydrocarbon lake margins (potentially biotic but non-aqueous);

and future red-giant ocean chemistry (water-based, aqueous). Each regime has different biosignature targets, and the *Dragonfly* mission primarily addresses the first two.

Following the framework of [Catling et al. \(2018\)](#), a credible biosignature requires context-dependence (interpretable against background abiotic chemistry), reproducibility, and resistance to false positives. Three measurement categories satisfy these criteria for Titan.

4.2.2 Enantiomeric Excess

Abiotic synthesis, including the Strecker reaction, produces racemic (equal L/D) amino acid mixtures. Terrestrial life produces exclusively L-amino acids. A statistically significant enantiomeric excess in DraMS amino acid measurements at Selk would be a strong biosignature. DraMS has demonstrated sensitivity to amino acid enantiomeric ratios for compounds expected at Titan ([Brinckerhoff et al., 2019](#)). However, small excesses ($<10\%$) can arise abiotically from asymmetric UV photodestruction or selective mineral surface adsorption; a biosignature interpretation requires excesses $\gg 10\%$, reproducibility across multiple traverse waypoints, and consistency with a self-replicating biochemical system.

4.2.3 Carbon Isotopic Fractionation

Biological carbon fixation preferentially incorporates ^{12}C over ^{13}C , producing $\delta^{13}\text{C}$ depletions of -25 to -30‰ (VPDB) in primary production organics on Earth. Abiotic tholin synthesis produces a distinct fractionation pattern driven by the kinetic isotope effect in UV photolysis. DraMS can measure organic carbon isotopic ratios; a $\delta^{13}\text{C}$ anomaly at the biological-range magnitude, reproducible at multiple sites along the traverse and correlated with the model’s highest-scoring terrain, would constitute a strong biosignature.

4.2.4 Acetylene Depletion

If acetylenotrophy (Equation 1.1) is active at the lake-margin environments identified as the globally highest-scoring in the present epoch, the surface C_2H_2 abundance should be measurably below photochemical model predictions (Vuitton et al., 2007). The anomalous downward H_2 flux reported by Strobel (2010) is consistent with this hypothesis. Yanez et al. (2024) estimated a predicted depletion of 5–15% below photochemical equilibrium for acetylenotrophy at a metabolic energy density comparable to the most energy-sparse known terrestrial organisms — within DraMS’s projected sensitivity. Crucially, the model identifies lake-margin terrain (not the Selk crater) as the highest-scoring environment for this biosignature, implying that a future mission directly targeting the north polar shorelines would be the most powerful test of the acetylenotrophy hypothesis.

4.2.5 Interpreting a Null Result

A null biosignature result at Selk would constrain specifically the “preserved impact-melt Strecker chemistry” pathway at that location and epoch. It would not constrain: (1) acetylenotrophy at the north polar lake margins, which score 15–20 percentage points higher than Selk in the present epoch and which *Dragonfly* does not visit; (2) life in the subsurface ocean, insensitive to surface sampling; or (3) extinct life from the LHB whose signatures may be buried below accessible depth. The habitability maps provide a rigorous framework for reporting a null result in terms of which specific pathways have been tested and which remain open.

4.3 Comparison with the Affholder et al. (2021)

Enceladus Framework

The closest methodological precedent for this work is Affholder et al. (2021), who applied Bayesian inference to Cassini INMS plume data from Enceladus to compare

the posterior probabilities of abiotic serpentinisation versus biological methanogenesis. Their conclusion — that the observed methane escape rate is most parsimoniously explained by methanogenesis if the prior probability of life exceeds ~ 0.35 — represents the most rigorous quantitative biosignature assessment published for any solar system body.

The present work extends the Affholder et al. approach in three dimensions. It is spatial, producing a 6.49-million-pixel posterior map rather than a single-point estimate. It is multi-feature, combining eight independent observational proxies rather than a single plume composition measurement. And it is multi-epoch, addressing three qualitatively distinct habitability regimes across geological time.

A key conceptual difference also warrants emphasis. [Affholder et al. \(2021\)](#) compute $P(\text{life} \mid \text{plume data})$ — the posterior probability that life exists, given chemical measurements — whilst this work computes $P(\text{habitable} \mid \mathbf{f})$ — the probability that the environment is suitable for life. This distinction, articulated by [Cockell \(2014\)](#) in the concept of “uninhabited but habitable” environments, matters practically: a habitable environment is a necessary but not sufficient condition for life. The present analysis identifies where to look; the Affholder-style framework tests, once measurements are made, whether life is actually present. The two approaches are complementary and sequential: habitability mapping prioritises targets, in-situ chemistry quantifies whether those targets are actually inhabited.

The recent results of [Affholder et al. \(2025\)](#) demonstrating thermodynamic viability of glycine fermentation in Titan’s subsurface ocean provide an important constraint on the subsurface ocean proximity feature (f_8). If the ocean is indeed habitable, then surface evidence of past subsurface–surface connectivity (the SAR-bright annuli at Selk and other crater sites) gains additional scientific weight. A future framework integrating the Affholder (2025) ocean habitability posterior with the present surface habitability maps, propagated through a physical model of subsurface–surface transport, would represent a significant advance.

4.4 Limitations of the Framework

4.4.1 Conditional Feature Independence

The Bayesian model treats the eight features as conditionally independent given the habitability latent variable H . This assumption simplifies the posterior update to a closed-form Beta distribution but introduces potential bias where features are correlated. In particular, organic abundance (f_2) and chemical energy (f_3) are positively correlated: both are elevated over tholin-rich organic terrain. Liquid hydrocarbon (f_1) and surface–atmosphere interaction (f_5) are structurally correlated through the lake-margin component of f_5 . Under positive correlation, the posterior is slightly over-estimated relative to a fully specified multivariate model. The magnitude of this bias can be estimated by comparing the independent-feature posterior with a correlated Gaussian copula model; preliminary assessment suggests the bias is <0.03 in posterior mean, smaller than the inter-epoch differences and the 94% HDI width. A full multivariate Bayesian model using a Gaussian copula structure is a natural extension for future work.

4.4.2 Static Geomorphological Template

The Lopes et al. (2019) geomorphological map is used for all five temporal configurations and all 72 interpolated epochs. In reality, the surface at -3.5 Gya had substantially different geomorphology: fewer confirmed dunes (the tholin inventory was smaller), different crater density (many present-day craters did not yet exist, and many LHB craters may have been subsequently buried or eroded), and no polar lakes. The future surface will be partially or completely submerged in the red-giant ocean. The use of the present-day map as a proxy for all epochs is an unavoidable simplification given the absence of independent temporal geomorphological information, and represents the largest qualitative uncertainty in the temporal analysis.

4.4.3 The Past-Epoch Cassini Contamination

As documented in Appendix D (A4), the PAST epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at -3.5 Gya. The resulting inflation of the north polar score ($P(H) \approx 0.70$, double the physically expected value) is mitigated in the multi-epoch reconstruction by using the Bayesian analytical formula for pre-lake-formation frames. However, the base-level inflation persists in the PAST static map. Readers should note that Table 3.3 reports crater scores from the PAST anchor, and these crater scores are not affected by the polar contamination (the impact-melt feature is spatially uncorrelated with the lake pattern). The north polar PAST score of ~ 0.70 is, however, an artefact and should be interpreted with caution.

4.5 Future Work

4.5.1 Post-Dragonfly Model Calibration

The most important near-term improvement to the framework will come from *Dragonfly* DraMS measurements at Selk crater and along the Belet–Selk traverse. Direct measurements of amino acid concentration and molecular complexity at each terrain class visited will calibrate the organic abundance scoring (f_2) at per-pixel resolution for dune, crater, and ejecta terrain types. These calibrated scores can be propagated globally via the Lopes (2019) terrain classification, reducing the dominant uncertainty in the organic abundance feature. A post-*Dragonfly* version of the framework, incorporating these calibration measurements, would produce substantially tighter posterior distributions globally and would directly improve the scientific value of any follow-on Titan orbital mission.

4.5.2 Extension to a Full Multivariate Model

The current Beta-conjugate model assumes feature independence. A natural extension is a multivariate Bayesian model using a Gaussian copula to represent feature correlations, estimated from the within-class co-variation in feature values across the global pixel population. The copula parameters can be estimated from the Cassini dataset directly, without requiring additional observations, and would reduce the positive bias associated with correlated positive features at lake margins. Implementation using PyMC or NumPyro is straightforward within the existing framework and would improve the credibility of the HDI estimates.

4.5.3 Additional Temporal Anchor Epochs

The current five-anchor temporal reconstruction has the largest uncertainties at the lake-formation transition (-1.5 to -0.5 Gya) and the solar-warming collapse ($+4.0$ to $+5.5$ Gya). These transitions are physically important — one represents the establishment of the present-day dominant habitability regime, the other its collapse — and are currently represented by interpolation between widely-spaced anchors. Adding three to five additional anchor epochs concentrated at these transitions, generated by running the full Bayesian pipeline with physically appropriate feature sets at those intermediate times, would substantially improve the temporal resolution and reduce the interpolation uncertainty.

4.6 Conclusions

This thesis has presented the first quantitative, spatially-resolved, multi-epoch Bayesian habitability map of Titan’s surface, derived from eight observational proxy features extracted from the complete Cassini remote sensing dataset. The principal conclusions are:

1. **North polar lake margins are the most habitable present-day environments on Titan’s surface.** The southern shoreline of Kraken Mare achieves

$P(H) \approx 0.58\text{--}0.63$, in the global top 3–5%. The dominant drivers are the coincidence of liquid hydrocarbon solvent, continuous fluvial delivery of organic material from the equatorial dune belt, and seasonal wetting–drying cycles that thermodynamically concentrate organic chemistry.

2. During the Late Heavy Bombardment, impact-melt ponds at crater sites were the primary habitability environments. Selk crater ranks highest among all confirmed impact structures at the past epoch ($P(H) = 0.225$ by the Bayesian formula), because its position at the organic-rich dune–plains boundary maximises Strecker amino acid synthesis potential in the melt.

3. The temporal habitability trajectory is non-monotonic. The transition from the present hydrocarbon-lake regime to the future red-giant ocean passes through a transient minimum at $\sim +5.0$ Gya, when the methane seas have evaporated but the surface is not yet warm enough for liquid water.

4. The red-giant ocean at +5.9 Gya represents an extraordinary future habitability scenario. A global water–ammonia ocean over a ~ 70 km accumulated organic sediment layer, maintained for ~ 0.8 Gyr, provides conditions for prebiotic chemistry of a richness unmatched elsewhere in the solar system.

5. The Dragonfly landing site is independently validated. Selk has the highest geomorphological diversity of any equatorial site ($7.8\times$ the global median), which directly corresponds to the terrain-boundary diversity cited in the mission science rationale. The highest-scoring features at Selk (geomorphological diversity and subsurface ocean proximity) directly correspond to the mission’s scientific rationale.

6. Feature importances validate the prior specification. All eight importance/weight ratios lie within 10% of unity, confirming that the geographic discrimination provided by each Cassini observable matches the astrobiological importance assigned through independent physical reasoning.

Together, these results demonstrate that Titan possesses a multilayered astrobiological significance extending from its geological past through the present to the far future, and that its habitability can be quantified spatially and temporally from the

Cassini legacy dataset with sufficient precision to guide mission planning and interpret in-situ measurements. The Bayesian multi-feature framework presented here provides a reproducible, extendable foundation for this work, and the directly testable predictions it makes for *Dragonfly* DraMS and DraGNS measurements at Selk represent the most important near-term validation pathway.

Bibliography

- Affholder, A. et al. (2025). Viability of glycine fermentation in Titan’s subsurface ocean. *The Planetary Science Journal*, 6:86.
- Affholder, A., Guyot, F., Sauterey, B., Ferrière, R., and Mazevet, S. (2021). Bayesian analysis of Enceladus’s plume data to assess methanogenesis. *Nature Astronomy*, 5:805–814.
- Artemieva, N. and Lunine, J. I. (2003). Cratering on Titan: impact melt, ejecta, and the fate of surface organics. *Icarus*, 164(2):471–480.
- Barnes, J. W. et al. (2007). Global-scale surface spectral variations on Titan seen from Cassini/VIMS. *Icarus*, 186(1):242–258.
- Benner, S. A., Ricardo, A., and Carrigan, M. A. (2004). Is there a common chemical model for life in the universe? *Current Opinion in Chemical Biology*, 8:672–689.
- Birch, S. P. D. et al. (2025). Titan’s volatile cycle. In *Titan After Cassini-Huygens*, chapter 10. Cambridge University Press.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45:5–32.
- Brinckerhoff, W. B., Grubisic, A., Danell, R. M., Kaplan, D., Getty, S., Li, X., et al. (2019). Dragonfly Mass Spectrometer (DraMS): Science overview and instrument status. *Bulletin of the American Astronomical Society*, 51:209. EPSC-DPS Joint Meeting 2019, abstract #609-12.
- Brown, M. E., Smith, A. L., Chen, C., and Ádámkovics, M. (2011). Discovery of fog at the south pole of Titan. *The Astrophysical Journal Letters*, 706:L110–L113.
- Brown, R. H. et al. (2008). The identification of liquid ethane in Titan’s Ontario Lacus. *Nature*, 454:607–610.

- Cable, M. L., Hörst, S. M., Hodyss, R., Beauchamp, P. M., Smith, M. A., and Willis, P. A. (2012). Titan tholins: simulating Titan organic chemistry in the Cassini-Huygens era. *Chemical Reviews*, 112:1882–1909.
- Catling, D. C., Krissansen-Totton, J., Kiang, N. Y., et al. (2018). Exoplanet biosignatures: A framework for their assessment. *Astrobiology*, 18(6):709–738.
- Cavicchioli, R. (2002). Extremophiles and the search for extraterrestrial life. *Astrobiology*, 2:281–292.
- Cockell, C. S. (2014). Trajectories of Martian habitability. *Astrobiology*, 14(2):182–203.
- Corlies, P., Hayes, A. G., Birch, S. P. D., Lorenz, R., Stiles, B. W., Kirk, R., and Poggiali, V. (2017). Titan’s topography and shape at the end of the cassini mission. *Geophysical Research Letters*, 44(23):11754–11761.
- Fulchignoni, M. et al. (2005). In situ measurements of the physical characteristics of Titan’s environment. *Nature*, 438:785–791.
- Gomes, R., Levison, H. F., Tsiganis, K., and Morbidelli, A. (2005). Origin of the cataclysmic Late Heavy Bombardment period of the terrestrial planets. *Nature*, 435:466–469.
- Grasset, O. and Pargamin, J. (2005). The ammonia–water system at high pressures: Implications for the methane of Titan. *Planetary and Space Science*, 53(4):371–384.
- Hayes, A., Aharonson, O., Callahan, P., et al. (2008). Hydrocarbon lakes on Titan: Distribution and interaction with a porous regolith. *Geophysical Research Letters*, 35(9):L09204.
- Hayes, A. G. (2016). The lakes and seas of Titan. *Annual Review of Earth and Planetary Sciences*, 44:57–83.
- Hedgepeth, J. E., Neish, C. D., Turtle, E. P., and Stiles, B. W. (2020). Titan’s impact crater population after Cassini. *Icarus*, 344:113664.

- Iess, L., Jacobson, R. A., Ducci, M., et al. (2012). The tides of Titan. *Science*, 337:457–459.
- Janssen, M. A. et al. (2016). Titan’s surface at 2.18-cm wavelength imaged by the Cassini RADAR radiometer. *Icarus*, 270:443–459.
- Jennings, D. E., Tokano, T., Cottini, V., Nixon, C. A., Achterberg, R. K., Flasar, F. M., Kunde, V. G., Romani, P. N., Samuelson, R. E., Segura, M. E., Gorius, N. J. P., Guandique, E., Kaelberer, M. S., and Coustenis, A. (2019). Titan surface temperatures during the Cassini mission. *The Astrophysical Journal Letters*, 877(1):L8.
- Lagouanelle, E., Beghin, C., Simoes, F., et al. (2024). Simulating Titan’s Schumann resonances to constrain the subsurface ocean depth with the Dragonfly EFIELD experiment. *Icarus*, 416:116073.
- Lavvas, P., Griffith, C. A., and Yelle, R. V. (2011). Condensation in Titan’s atmosphere at the Huygens landing site. *Icarus*, 215:732–750.
- Le Mouélic, S. et al. (2019). The Cassini VIMS archive of Titan: from browse products to global infrared color maps. *Icarus*, 319:121–132.
- Lopes, R. M. C. et al. (2016). Titan as revealed by the Cassini radar. *Space Science Reviews*, 215:33.
- Lopes, R. M. C. et al. (2020). A close look at Titan’s geology. *Journal of Geophysical Research: Planets*.
- Lopes, R. M. C., Kirk, R. L., Mitchell, K. L., et al. (2013). Cryovolcanism on Titan: New results from Cassini RADAR and VIMS. *Journal of Geophysical Research: Planets*, 118:416–435.
- Lopes, R. M. C., Malaska, M. J., Schoenfeld, A. M., et al. (2019). A global geomorphologic map of Saturn’s moon Titan. *Nature Astronomy*, 4:228–233.

- Lorenz, R. D. (2014). The flushing of Ligeia: Composition variations across Titan’s seas in a simple diffusion model. *Geophysical Research Letters*, 41(15):5764–5770.
- Lorenz, R. D. et al. (2006). The sand seas of Titan: Cassini RADAR observations of longitudinal dunes. *Science*, 312:724–727.
- Lorenz, R. D. et al. (2008). Titan’s inventory of organic surface materials. *Geophysical Research Letters*, 35:L02206.
- Lorenz, R. D. et al. (2018). Dragonfly: A rotorcraft lander concept for scientific exploration at Titan. *Johns Hopkins APL Technical Digest*, 34(3):374–387.
- Lorenz, R. D., Lunine, J. I., and McKay, C. P. (1997). Titan under a red giant sun: A new kind of “habitable” moon. *Geophysical Research Letters*, 24(22):2905–2908.
- Lorenz, R. D., MacKenzie, S. M., Neish, C. D., Le Gall, A., Turtle, E. P., Barnes, J. W., Trainer, M. G., Werynski, A., Hedgepeth, J. E., and Karkoschka, E. (2021). Selection and characteristics of the Dragonfly landing site near Selk crater, Titan. *The Planetary Science Journal*, 2(1):24.
- Madan, R. et al. (2026). Amino acid synthesis in Titan impact-melt environments. *Icarus*. In press / preprint; DOI to be confirmed at publication.
- Malaska, M. J. et al. (2020). Labyrinth terrain on Titan. *Icarus*, 344:113764.
- Malaska, M. J. et al. (2025). Global geology of Titan. In *Titan After Cassini-Huygens*, chapter 9. Cambridge University Press.
- Mastrogiuseppe, M., Hayes, A. G., Poggiali, V., et al. (2019). Deep and methane-rich lakes on Titan. *Nature Astronomy*, 3:535–542.
- Mastrogiuseppe, M., Poggiali, V., Hayes, A., Lorenz, R., Lunine, J., Picardi, G., Seu, R., Flamini, E., Mitri, G., Notarnicola, C., Paillou, P., and Zebker, H. (2014). The bathymetry of a Titan sea. *Geophysical Research Letters*, 41(5):1432–1437.
- McKay, C. P. (2016). Titan as the abode of life. *Life*, 6(1):8.

- McKay, C. P. and Smith, H. D. (2005). Possibilities for methanogenic life in the environments of Titan. *Icarus*, 178(1):274–276.
- Meyer-Dombard, D. R. et al. (2025). Habitability of Titan’s interior and surface. In *Titan After Cassini-Huygens*, chapter 14. Cambridge University Press.
- Miller, J., Hayes, A. G., Corlies, P., Birch, S. P. D., Lorenz, R. D., and Turtle, E. P. (2021). Global fluvial channel map of Titan from Cassini SAR. *Journal of Geophysical Research: Planets*, 126(12).
- Mitchell, J. L. and Lora, J. M. (2016). The climate of Titan. *Annual Review of Earth and Planetary Sciences*, 44:353–380.
- Mitri, G., Meriggiola, R., Hayes, A., et al. (2014). Shape, surface topography, and tectonic deformation of Titan. *Icarus*, 236:169–177.
- Neish, C. D., Barnes, J. W., Sotin, C., et al. (2018). Titan as a prebiotic chemical system. *Astrobiology*, 18(5):571–585.
- Neish, C. D., Blichert-Toft, J., et al. (2024). Evidence for ancient subsurface ocean–surface exchange on Titan from crater ejecta composition. *Nature Geoscience*, 17:238–244.
- Neish, C. D., Somogyi, Á., Lunine, J. I., and Smith, M. A. (2009). Low temperature hydrolysis of laboratory tholins in ammonia–water solutions: implications for prebiotic chemistry on Titan. *Icarus*, 201:412–421.
- Neish, C. D., Somogyi, Á., and Smith, M. A. (2010). Titan’s primordial soup: Formation of amino acids via low-temperature hydrolysis of Tholins. *Astrobiology*, 10(3):337–347.
- Niemann, H. B., Atreya, S. K., Bauer, S. J., et al. (2005). The abundances of constituents of Titan’s atmosphere from the GCMS instrument on the Huygens probe. *Nature*, 438:779–784.

- Nimmo, F. and Bills, B. G. (2010). Shell thickness variations and the long-wavelength topography of Titan. *Icarus*, 208:896–904.
- O’Brien, D. P., Lorenz, R. D., and Lunine, J. I. (2005). Numerical calculations of the longevity of impact oases on Titan. *Icarus*, 173:243–253.
- Palmer, M. Y., Cordiner, M. A., Nixon, C. A., Charnley, S. B., Teanby, N. A., Kisiel, Z., Irwin, P. G. J., and Mumma, M. J. (2017). ALMA detection and astrobiological potential of vinyl cyanide on Titan. *Science Advances*, 3(7):e1700022.
- Pappalardo, R. T. et al. (1999). Does Europa have a subsurface ocean? evaluation of the geological evidence. *Journal of Geophysical Research: Planets*, 104:24015–24055.
- Pizzarello, S. (2006). The chemistry of life’s origin: A carbonaceous meteorite perspective. *Accounts of Chemical Research*, 39(4):231–237.
- Radebaugh, J., Lorenz, R. D., Farr, T., Paillou, P., Savage, C., and Spencer, C. (2011). Linear dunes on Titan and earth: initial remote sensing comparisons. *Geomorphology*, 121:122–132.
- Rothschild, L. J. and Mancinelli, R. L. (2001). Life in extreme environments. *Nature*, 409:1092–1101.
- Sagan, C. and Khare, B. N. (1979). Tholins: organic chemistry of interstellar grains and gas. *Nature*, 277:102–107.
- Schröder, K.-P. and Connors Smith, R. (2008). Distant future of the Sun and Earth revisited. *Monthly Notices of the Royal Astronomical Society*, 386:155–163.
- Seignovert, B., Le Mouélic, S., Brown, R. H., Karkoschka, E., Pasek, V., Sotin, C., and Turtle, E. P. (2019). Titan’s global map combining VIMS and ISS mosaics.
- Shannon, C. E. (1948). A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423.

- Shock, E. L. and Holland, M. E. (2007). Quantitative habitability. *Astrobiology*, 7(6):839–851.
- Solomonidou, A. et al. (2018). The spectral nature of Titan’s major geomorphological units: constraints on surface composition. *Journal of Geophysical Research: Planets*, 123:489–507.
- Stevenson, J., Lunine, J., and Clancy, P. (2015). Membrane alternatives in worlds without oxygen: creation of an azotosome. *Science Advances*, 1(1):e1400067.
- Stofan, E. R., Elachi, C., Lunine, J. I., et al. (2007). The lakes of Titan. *Nature*, 445:61–64.
- Strobel, D. F. (2010). Molecular hydrogen in Titan’s atmosphere: Implications of the Cassini INMS measurements. *Icarus*, 208(2):878–886.
- Teolis, B. D. et al. (2015). A revised sensitivity model for Cassini CAPS-ELS: assessment and application. *Journal of Geophysical Research: Space Physics*, 120:7124–7138.
- Tobie, G., Grasset, O., Lunine, J. I., Mocquet, A., and Sotin, C. (2005). Titan’s internal structure inferred from a coupled thermal-orbital model. *Icarus*, 175:496–502.
- Tokano, T. (2010). Relevance of fast westerlies at equinox for the eastward elongation of Titan’s dunes. *Aeolian Research*, 2:113–127.
- Tosi, N., Kalousová, K., Spohn, T., et al. (2022). Science goals and new mission concepts for future exploration of Titan’s atmosphere, geology and habitability: POSEIDON (Titan Polar Scout/Orbiter and In Situ Lake Lander and Drone Explorer). *Experimental Astronomy*, 54:1117–1177.
- Turtle, E. P., Perry, J. E., Hayes, A. G., et al. (2011a). Rapid and pervasive changes in surface longevity and coverage observed by Cassini VIMS: Titan’s active methane cycle. *Science*, 331:1414–1417.

- Turtle, E. P., Perry, J. E., Hayes, A. G., Lorenz, R. D., Barnes, J. W., McEwen, A. S., West, R. A., Del Genio, A. D., Barbara, J. M., Lunine, J. I., Schaller, E. L., Ray, T. L., Lopes, R. M. C., and Stofan, E. R. (2011b). Shoreline retreat at Titan’s Ontario Lacus and Arrakis Planitia from Cassini imaging science subsystem observations. *Icarus*, 212(2):957–959.
- Turtle, E. P., Perry, J. E., McEwen, A. S., et al. (2009). Cassini imaging of Titan’s high-latitude lakes, clouds, and south-polar surface changes. *Geophysical Research Letters*, 36:L02204.
- Tyler, G. L., Eshleman, V. R., Anderson, J. D., et al. (1981). Radio science investigations of the Saturn system with Voyager 1: Preliminary results. *Science*, 212:201–206.
- Vuitton, V., Yelle, R. V., and McEwan, M. J. (2007). Ion chemistry and N-containing molecules in Titan’s upper atmosphere. *Icarus*, 191:722–742.
- Waite, J. H. et al. (2007). The process of tholin formation in Titan’s upper atmosphere. *Science*, 316:870–875.
- Waite, J. H. et al. (2017). Cassini finds molecular hydrogen in the Enceladus plume: evidence for hydrothermal processes. *Science*, 356:155–159.
- Wood, C. A. et al. (2010). Impact craters on Titan. *Icarus*, 206:334–344.
- Yanez, J. et al. (2024). Acetylenotrophy and the habitability of Titan’s surface. *Icarus*, 408:115969.
- Zebker, H. A., Stiles, B., Hensley, S., et al. (2009). Size and shape of Saturn’s moon Titan. *Science*, 324(5929):921–923.

A Physical Parameters for Titan

Table A.1: Physical constants and observationally-constrained parameters for Titan used throughout this thesis.

Parameter	Value	Reference
Mean radius	2575.0 km	IAU 2015
Surface gravity	1.352 m s^{-2}	Zebker et al. (2009)
Surface pressure	$1.467 \pm 0.001 \text{ bar}$	Fulchignoni et al. (2005)
Surface temperature	$93.65 \pm 0.25 \text{ K}$	Fulchignoni et al. (2005)
N ₂ fraction (surface)	$\approx 94.2\%$	Niemann et al. (2005)
CH ₄ fraction (surface)	$5.65 \pm 0.18\%$	Niemann et al. (2005)
Tidal Love number k_2	0.589 ± 0.150	Iess et al. (2012)
Saturn orbital period	29.46 yr	—
Axial obliquity	26.73°	—
Solar flux at Titan	$\approx 14.9 \text{ W m}^{-2}$	at 9.54 AU
Organic deposition rate	$\sim 5 \times 10^{-14} \text{ g cm}^{-2} \text{ s}^{-1}$	Lavvas et al. (2011)
North polar sea volume	$\sim 70,000 \text{ km}^3$	Mastrogiuseppe et al. (2019)
Water–ammonia eutectic	176 K at 32.6 wt% NH ₃	Grasset and Pargamin (2005)
Subsurface ocean depth	$\sim 55\text{--}80 \text{ km}$	Tobie et al. (2005)

B Bayesian Prior Specification

Table B.1: Prior means μ_i and weights w_i for all eight habitability features at the present (Cassini) epoch. Global prior mean $\mu_0 = \sum_i w_i \mu_i = 0.331$. Concentration parameter $\kappa = 5$ is applied uniformly.

Feature	Symbol	w_i	μ_i	Key ref.
Liquid hydrocarbon	f_1	0.25	0.020	Hayes et al. (2008)
Organic abundance	f_2	0.20	0.700	Malaska et al. (2025)
Chemical energy	f_3	0.20	0.350	Strobel (2010)
Methane cycle	f_4	0.15	0.400	Mitchell and Lora (2016)
Surface–atm. interaction	f_5	0.08	0.350	Miller et al. (2021)
Topographic complexity	f_6	0.06	0.250	Corlies et al. (2017)
Geomorphological diversity	f_7	0.04	0.300	Lopes et al. (2019)
Subsurface ocean proximity	f_8	0.02	0.030	Iess et al. (2012)
<i>Global prior mean μ_0</i>			0.331	

C Geomorphological Organic Abundance Scores

Table C.1: Organic abundance proxy scores used for the geomorphological seamless mode of Feature 2. Grounded in published VIMS spectral characterisation of each terrain class.

Class	Name	Physical basis	Score
Dunes	Equatorial sand seas	Tholin-dominated; highest VIMS ratio (Lorenz et al., 2006)	0.82
Plains	Smooth lowlands	Organic sediment blanket (Lopes et al., 2016)	0.68
Basins	Topographic lows	Organic accumulation (Lopes et al., 2020)	0.58
Labyrinth	Dissected terrain	Spectral affinity with plains (Malaska et al., 2020)	0.55
Craters	Impact structures	Mixed water-ice and organic ejecta (Neish et al., 2018)	0.35
Mountains	Ridge terrain	Water-ice-rich crustal material (Solomonidou et al., 2018)	0.25
Lakes	Hydrocarbon seas	Liquid hydrocarbons; low solid organic content (Brown et al., 2008)	0.05

D Declared Assumptions and Known Limitations

- A1. Conditional feature independence.** The eight features are treated as conditionally independent given habitability H . In practice, organic abundance and chemical energy are positively correlated, as are liquid hydrocarbon and surface-atmosphere interaction. The independence assumption is conservative: correlated positive features slightly overstate the posterior, but the framework remains physically motivated and the data-derived importances are consistent with the prior weights (Section 3.2.5).
- A2. Static geomorphological template.** The Lopes et al. (2019) map is used for all five temporal configurations. The Late Heavy Bombardment surface had different dune distributions and crater density, and the future surface will be submerged. The present template is used as the best available proxy, with amplitude modifications applied through epoch-specific prior means and scale functions.
- A3. Seamless organic abundance mode.** The organic abundance feature uses terrain-class scores globally rather than VIMS spectral band ratios, because the VIMS mosaic covers only $\sim 50\%$ of the globe and direct blending produced a visible seam at the coverage boundary at 180°W .
- A4. Past-epoch Cassini feature contamination.** The past-epoch posterior was computed using present-era Cassini SAR maps, which encode the present polar lake morphology even at $t = -3.5$ Gya. The north polar region is consequently inflated to $P(H) \approx 0.70$ in the past anchor, approximately double the physically expected value. This is mitigated in the multi-epoch interpolation and documented explicitly in the Results.

A5. Multi-epoch interpolation accuracy. PCHIP interpolation between anchor posteriors is not equivalent to independent epoch-specific inference. Intermediate maps correctly reproduce the anchors and are monotonic between them, but spatial patterns reflect blended anchor structures. Estimated uncertainty: $\Delta P(H) \lesssim \pm 0.05$.

A6. SAR coverage gap. Approximately 33% of Titan’s surface lacks SAR coverage. Features 1, 3, and 8 are degraded in these regions, where geomorphological and topographic proxies substitute.